On the precision of dynamic gene regulation

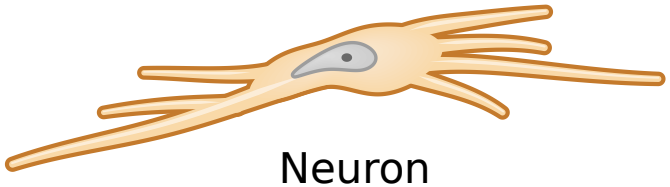
Tarek Tohme

Under the supervision of

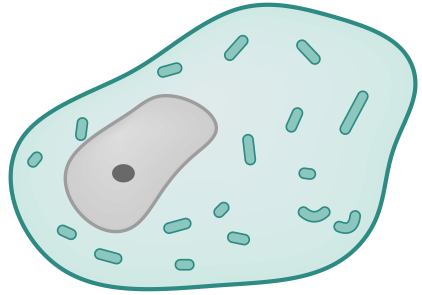
Aleksandra Walczak and Thierry Mora

November 4, 2026



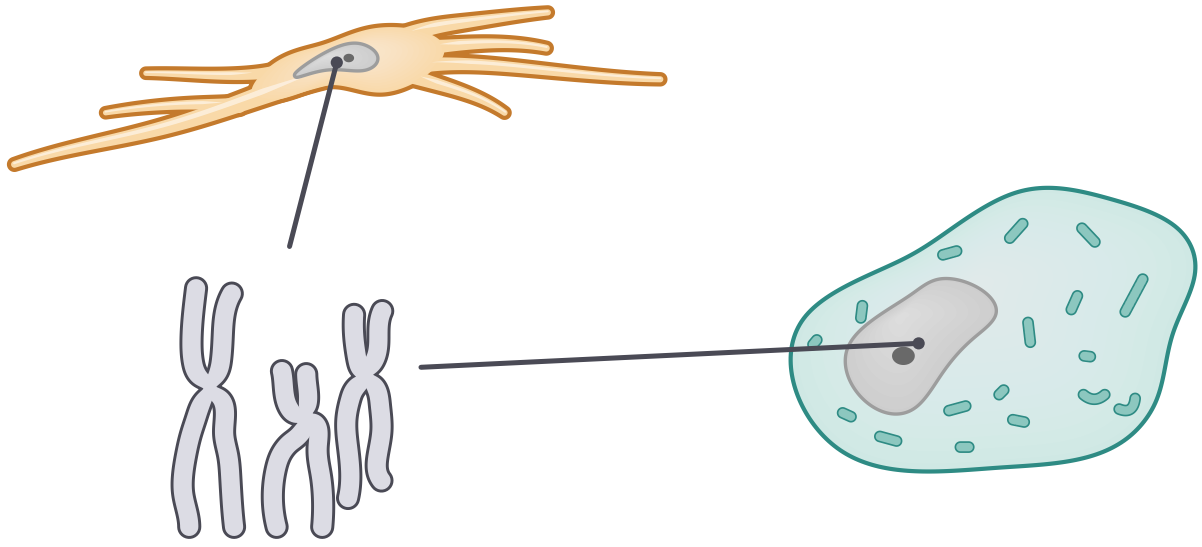


Neuron



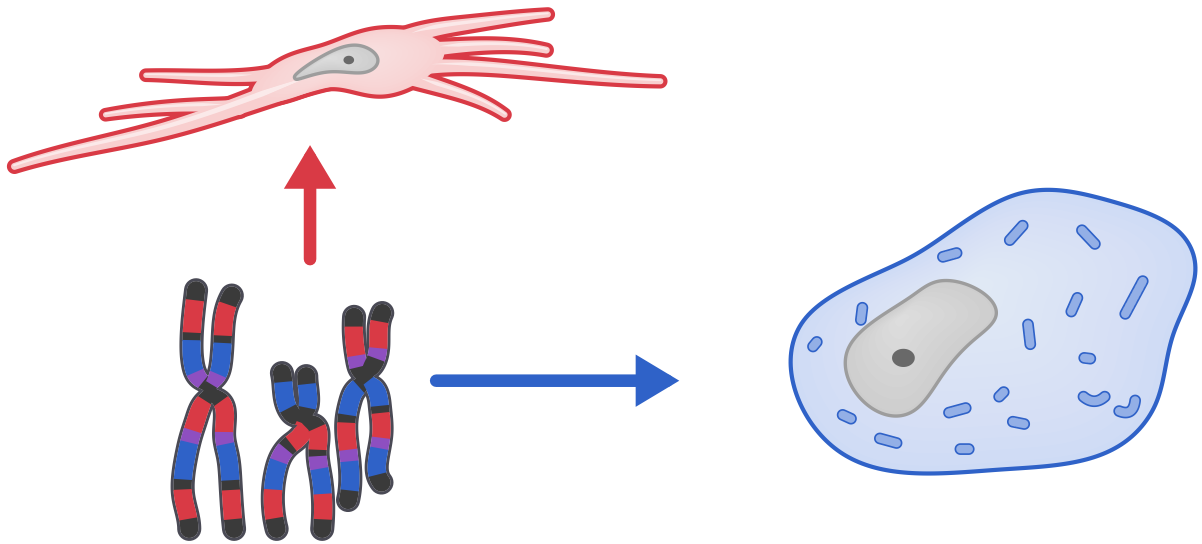
Kidney cell

scene1



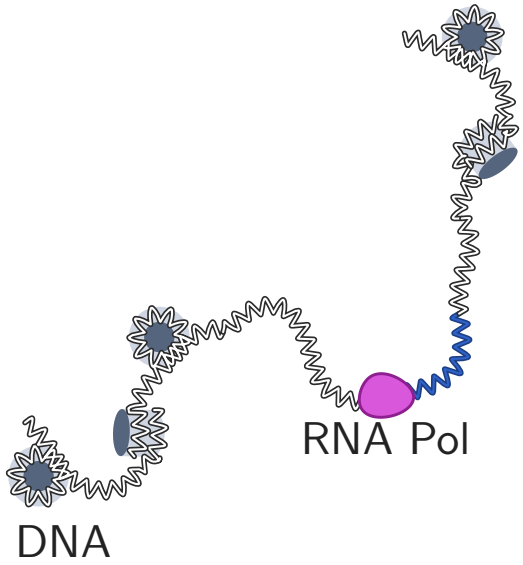
scene2

Gene expression is regulated



scene3

Overview of gene regulation

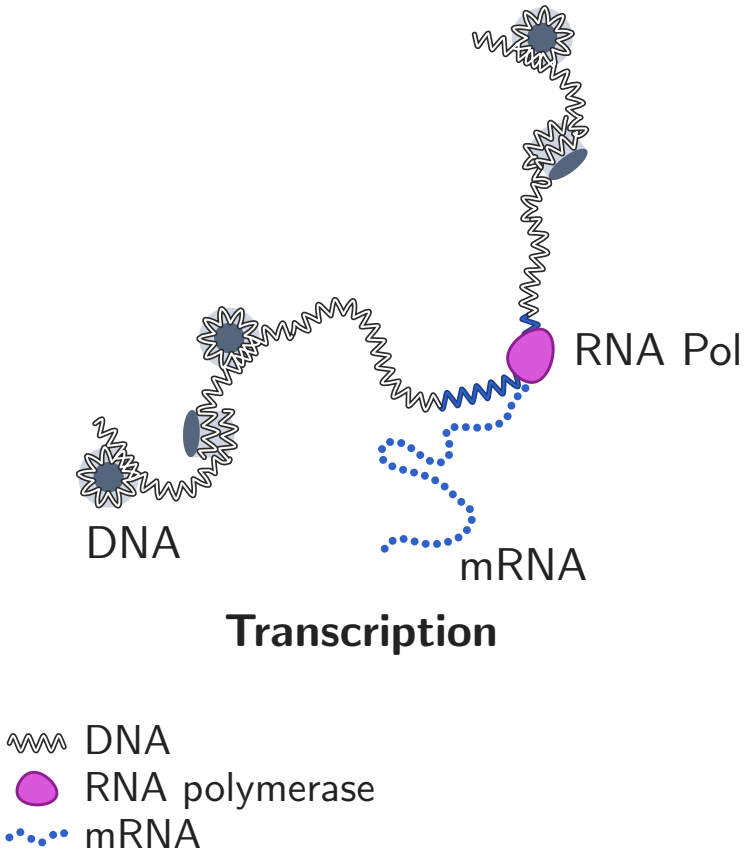


Transcription

DNA
RNA polymerase

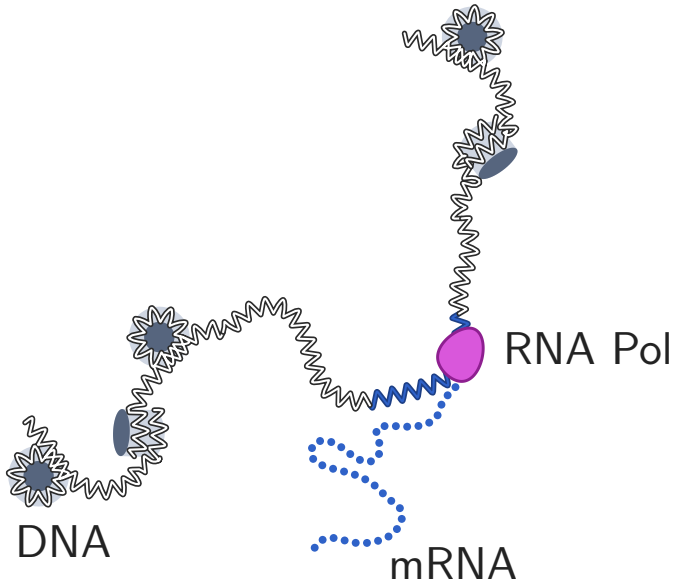
scene4_1

Overview of gene regulation

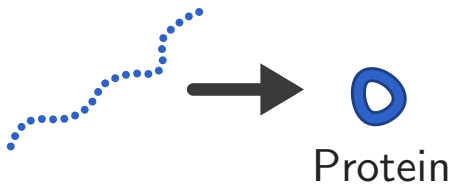


scene4_2

Overview of gene regulation



Transcription

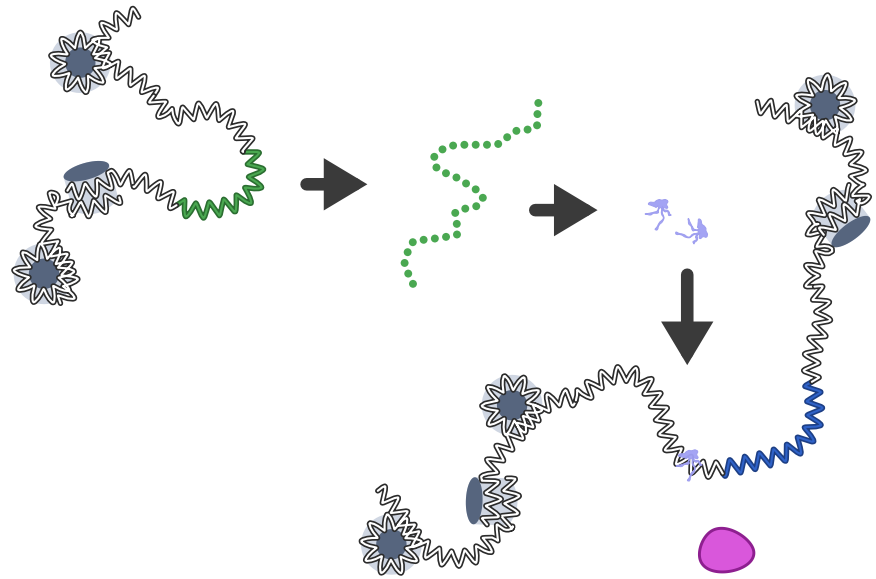


Translation

DNA
RNA polymerase
mRNA

scene4$_3$

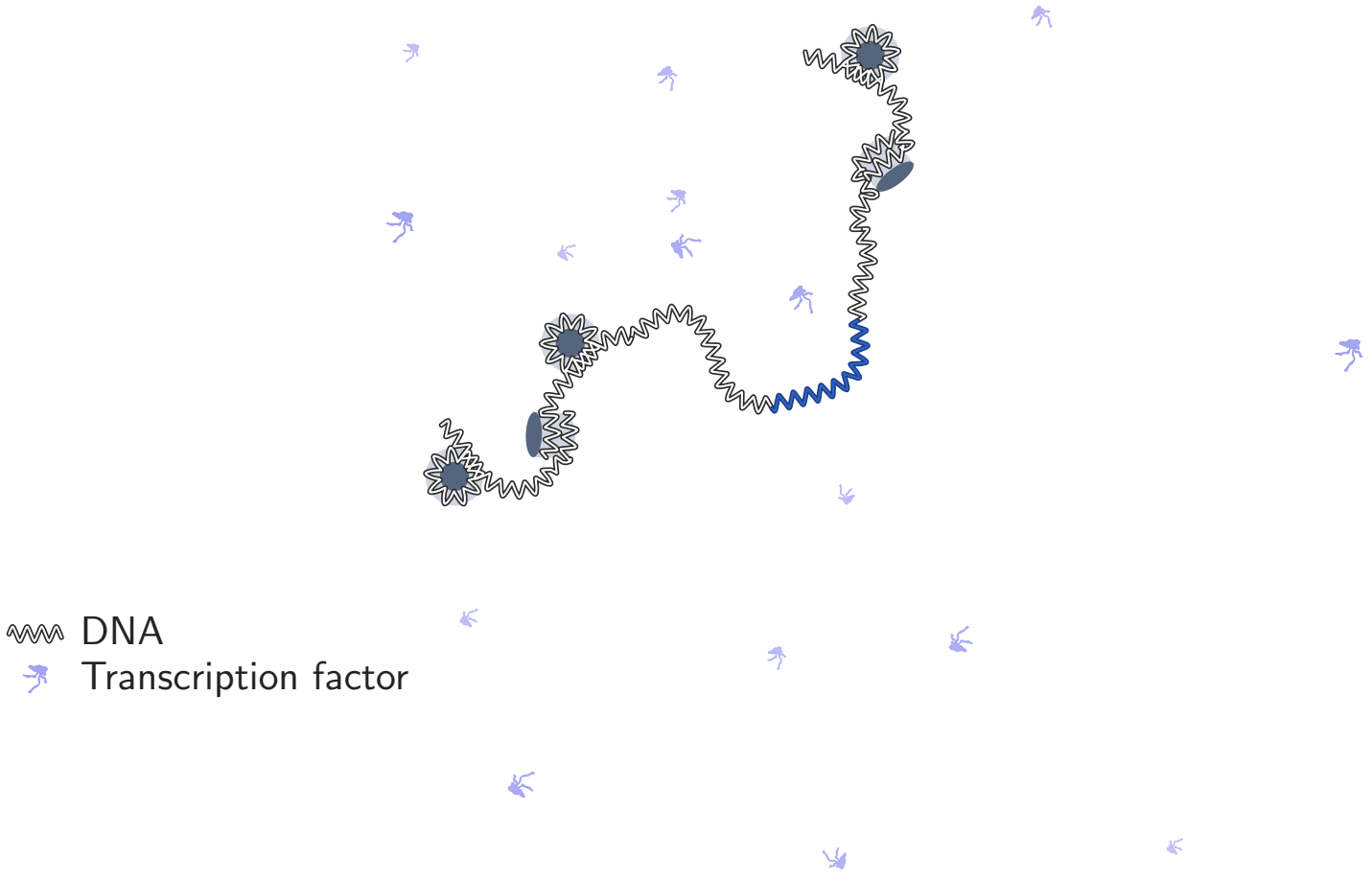
Overview of gene regulation



DNA
Transcription factor
RNA polymerase
mRNA

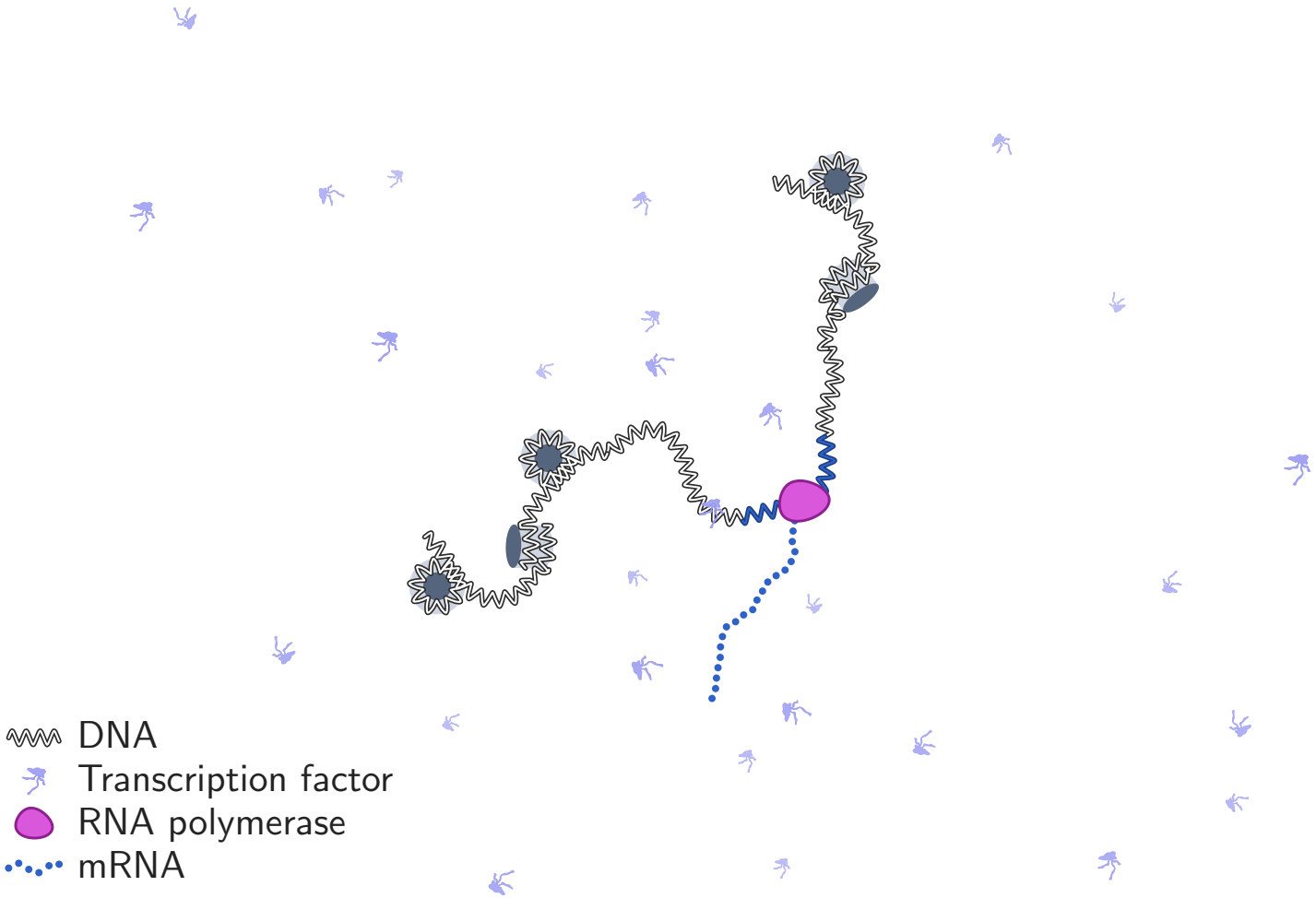
scene5

Expression levels are continuous ...



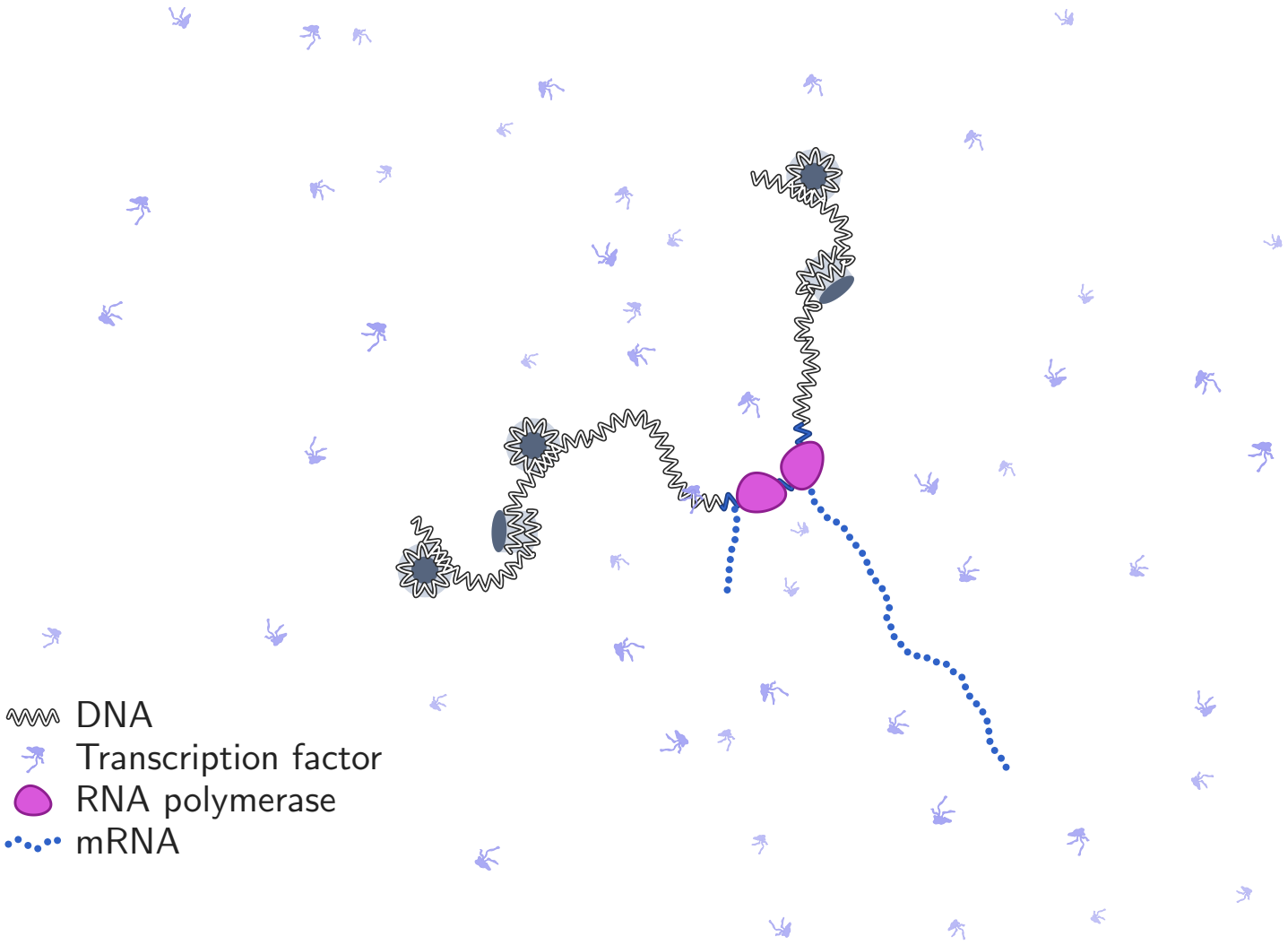
scene6$_1$

Expression levels are continuous ...



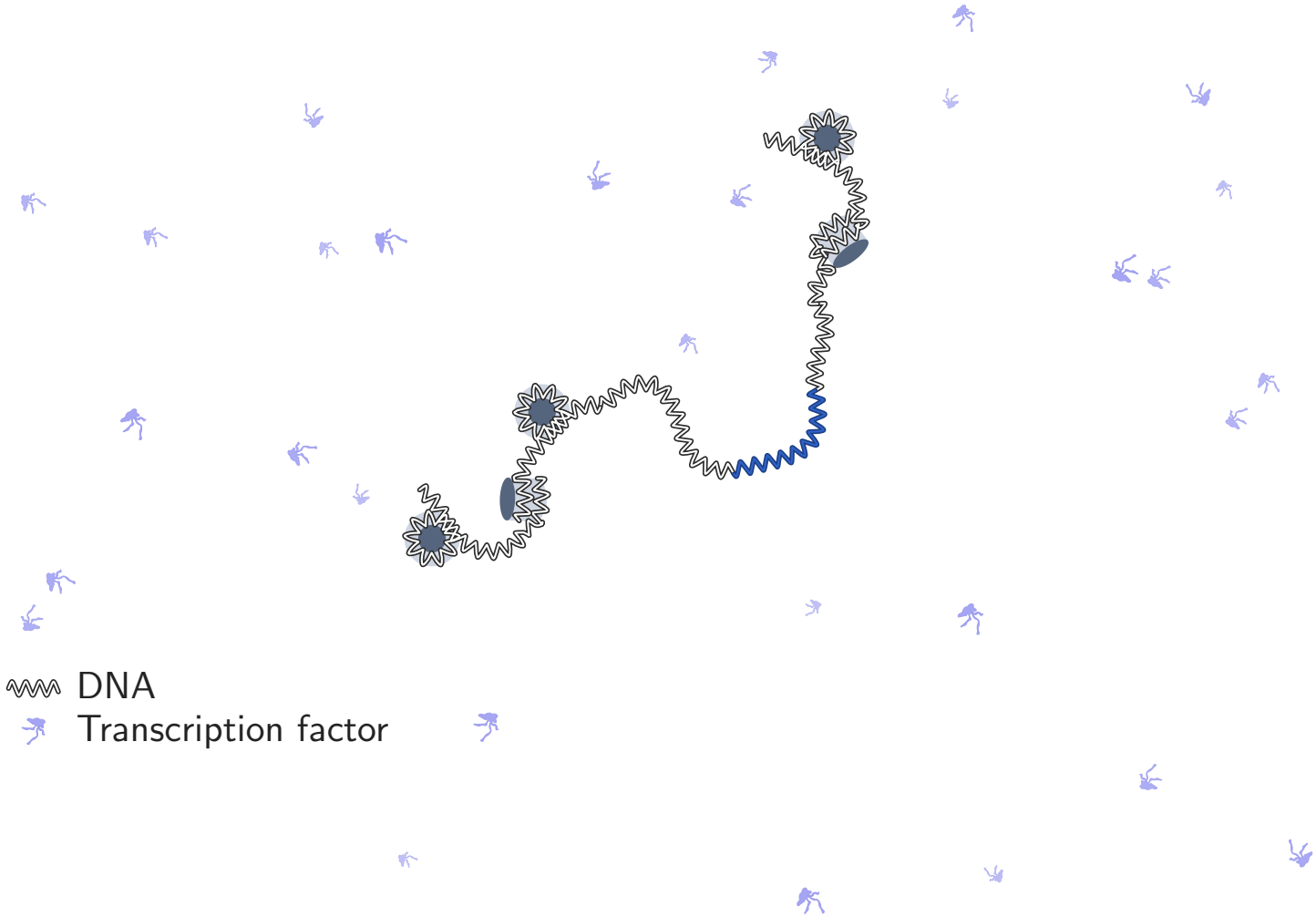
scene6_2

Expression levels are continuous ...



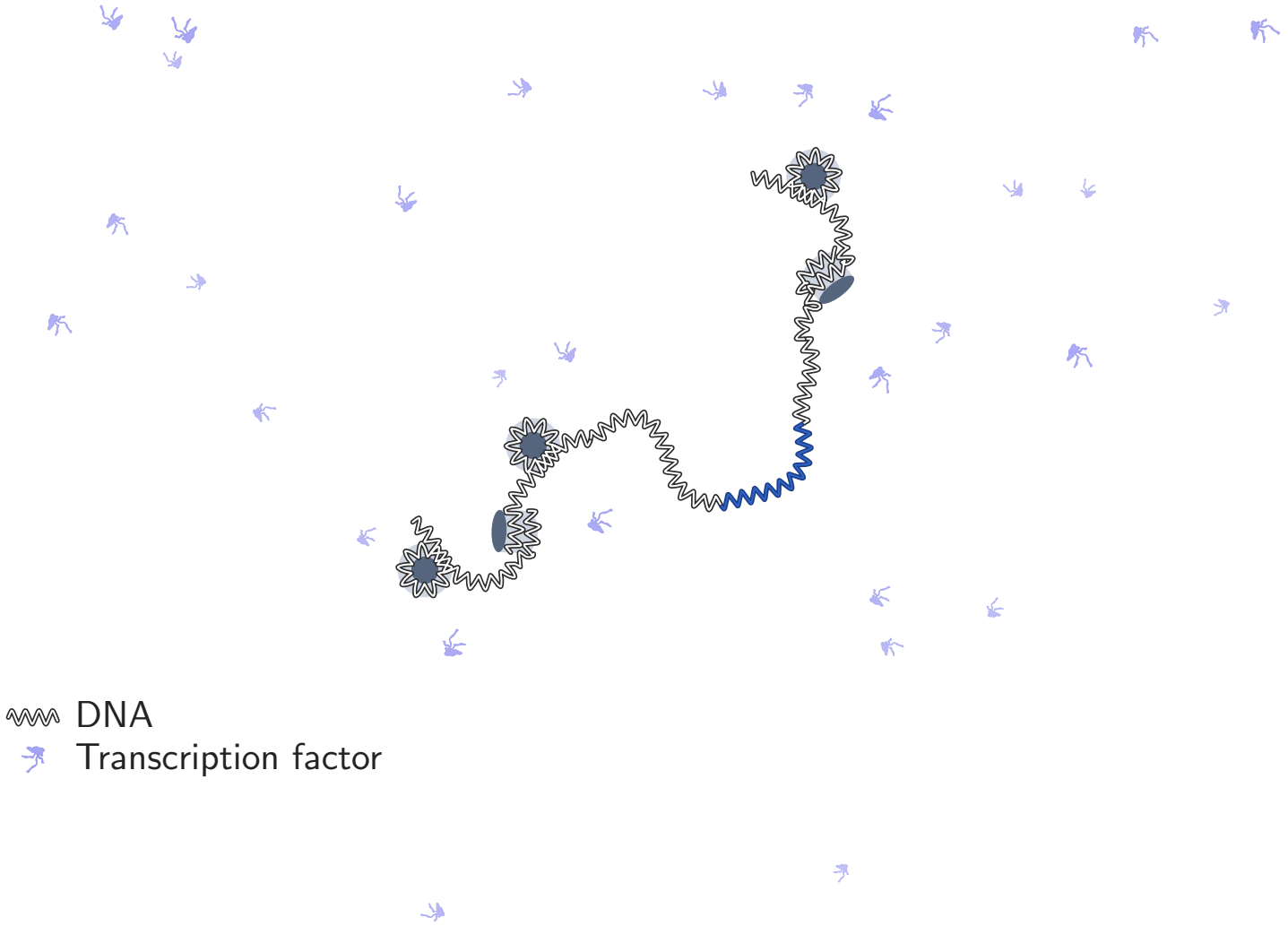
scene6$_3$

... but transcriptional control is noisy



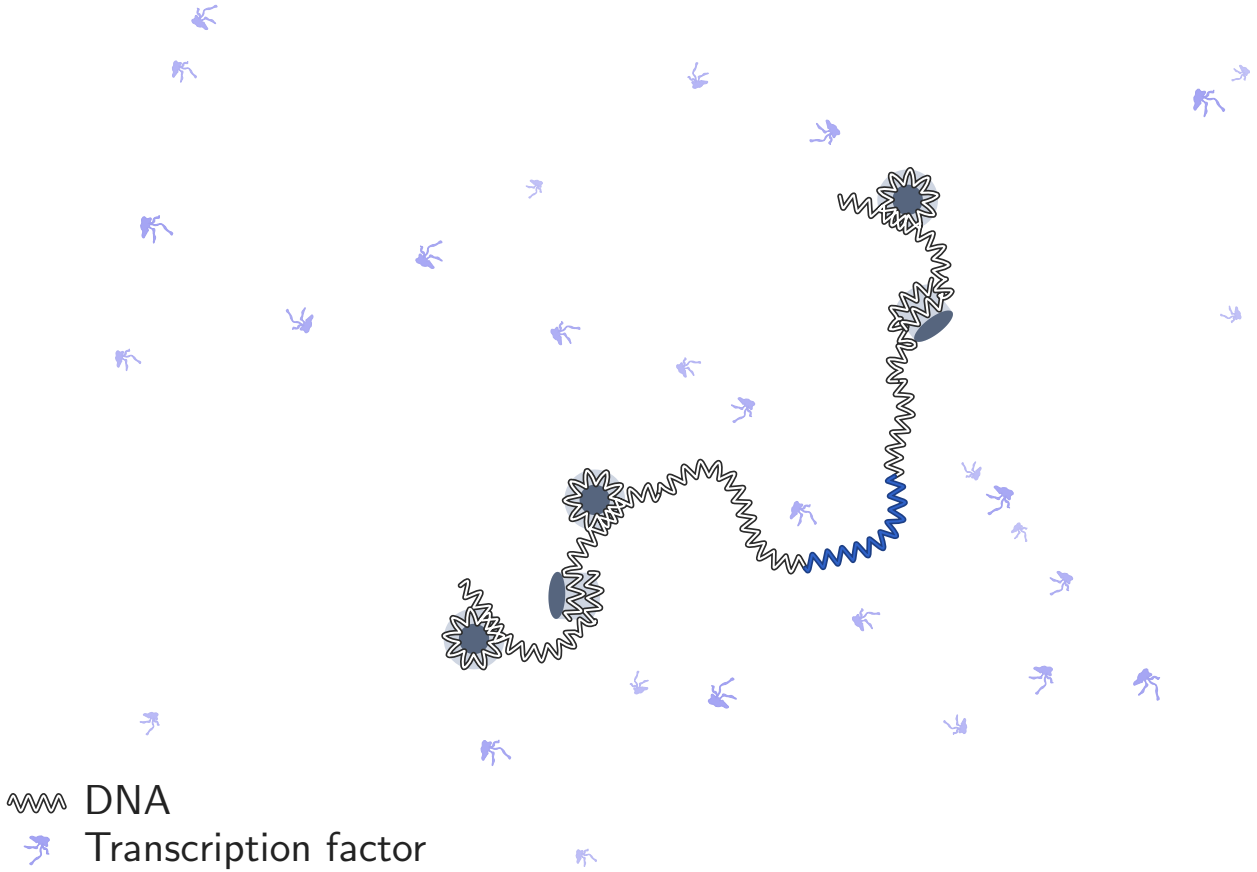
scene6$_4$

... but transcriptional control is noisy



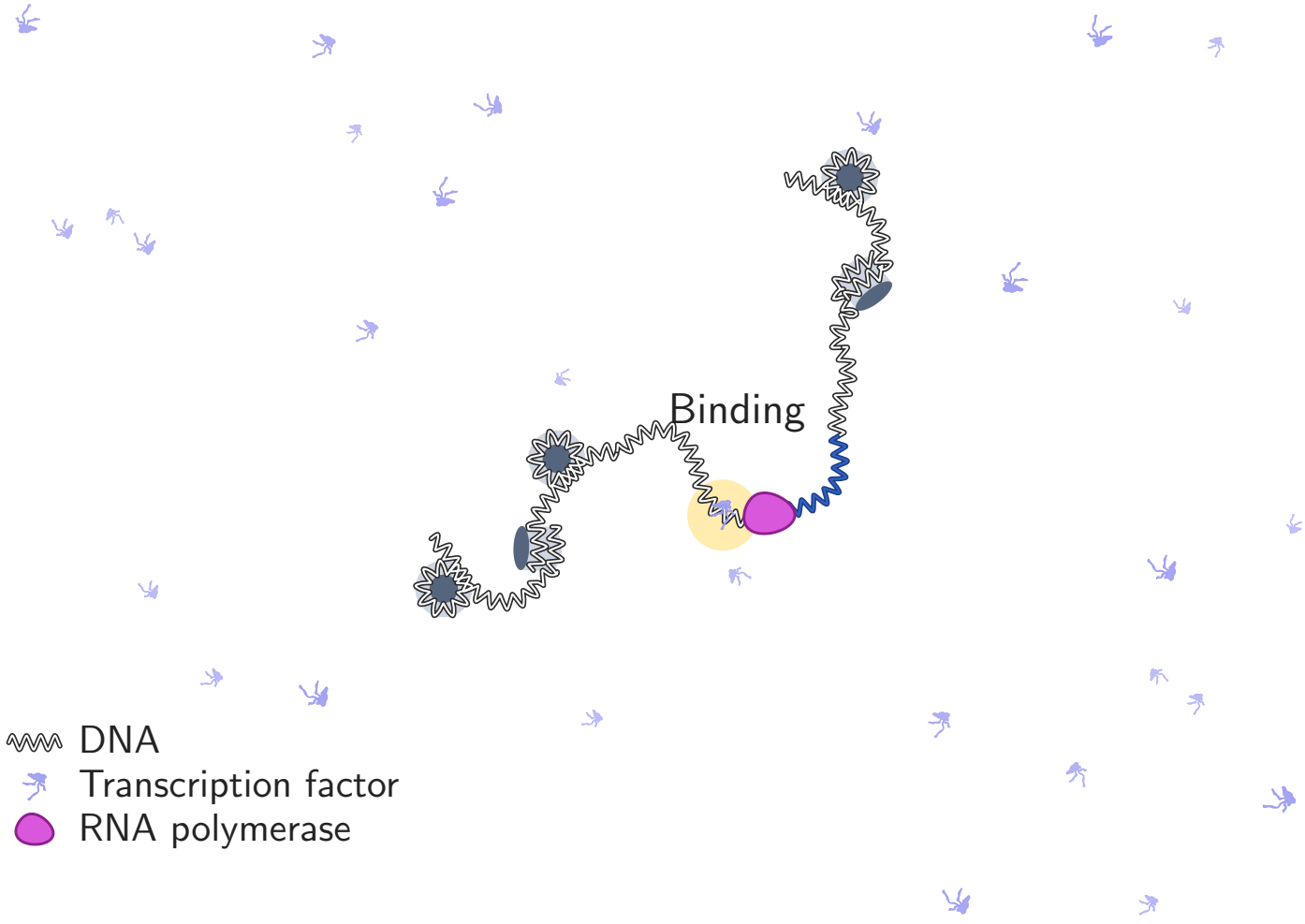
scene6_5

... but transcriptional control is noisy



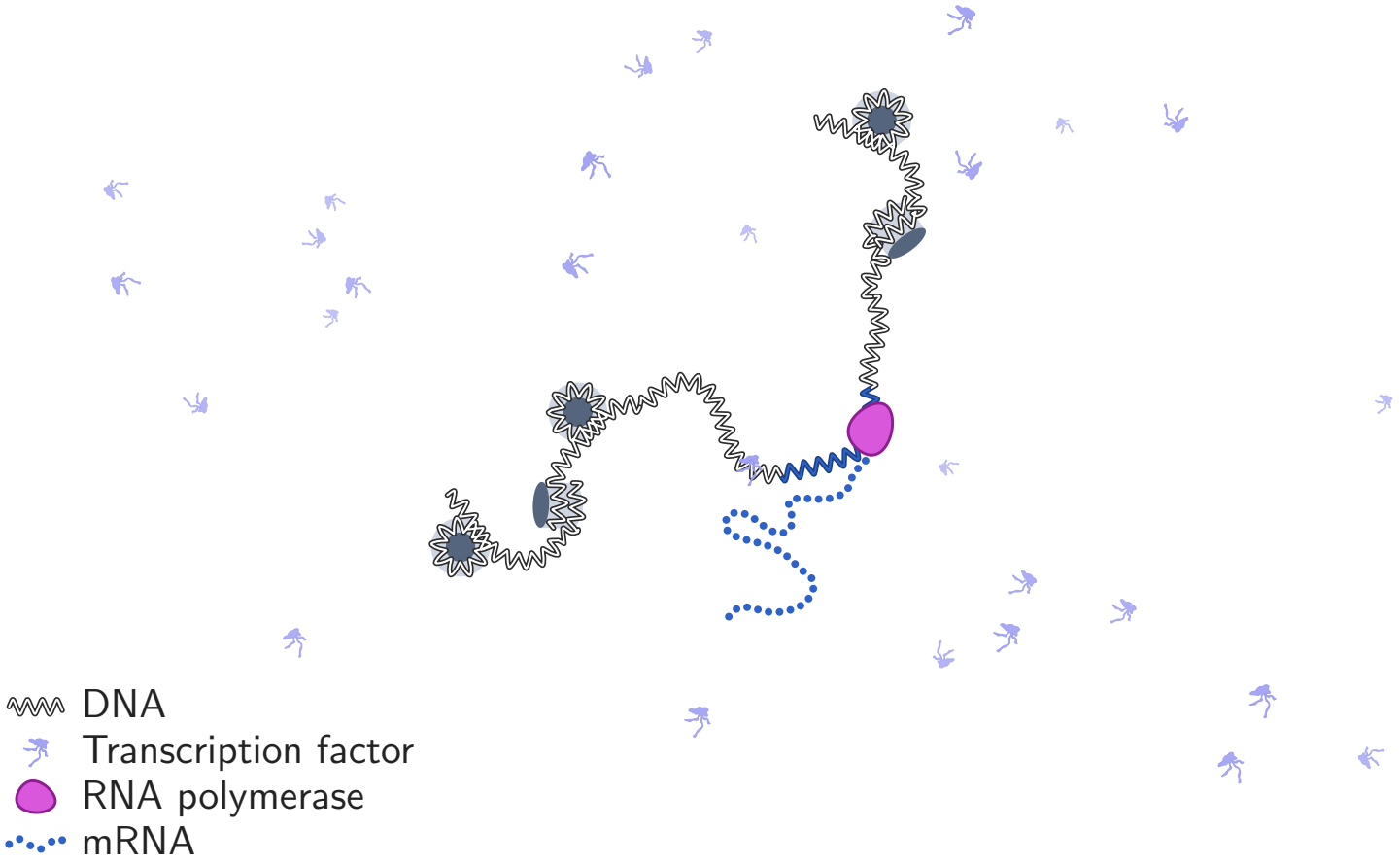
scene6_6

... but transcriptional control is noisy



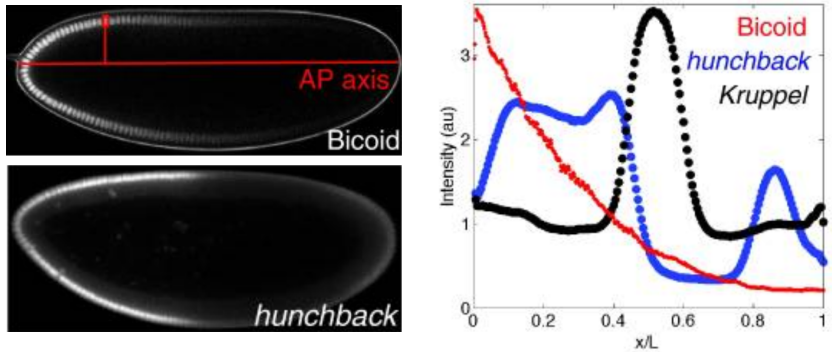
scene6$_7$

... but transcriptional control is noisy



scene6$_8$

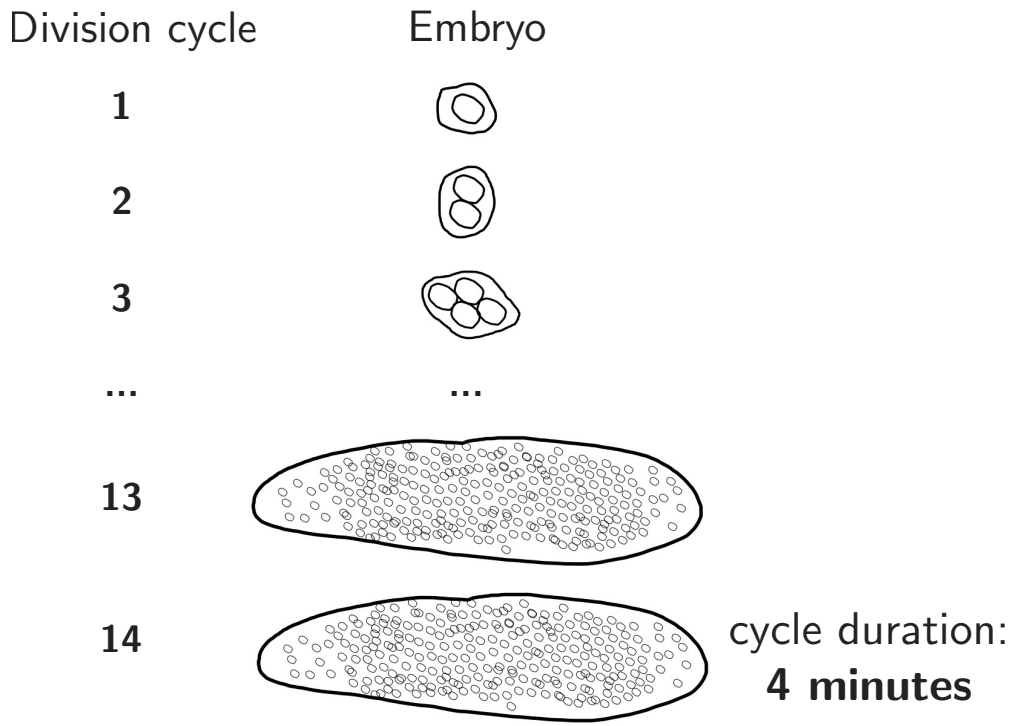
Sensitive response to shallow concentration gradients



Grimm et al., *Development,* 2010

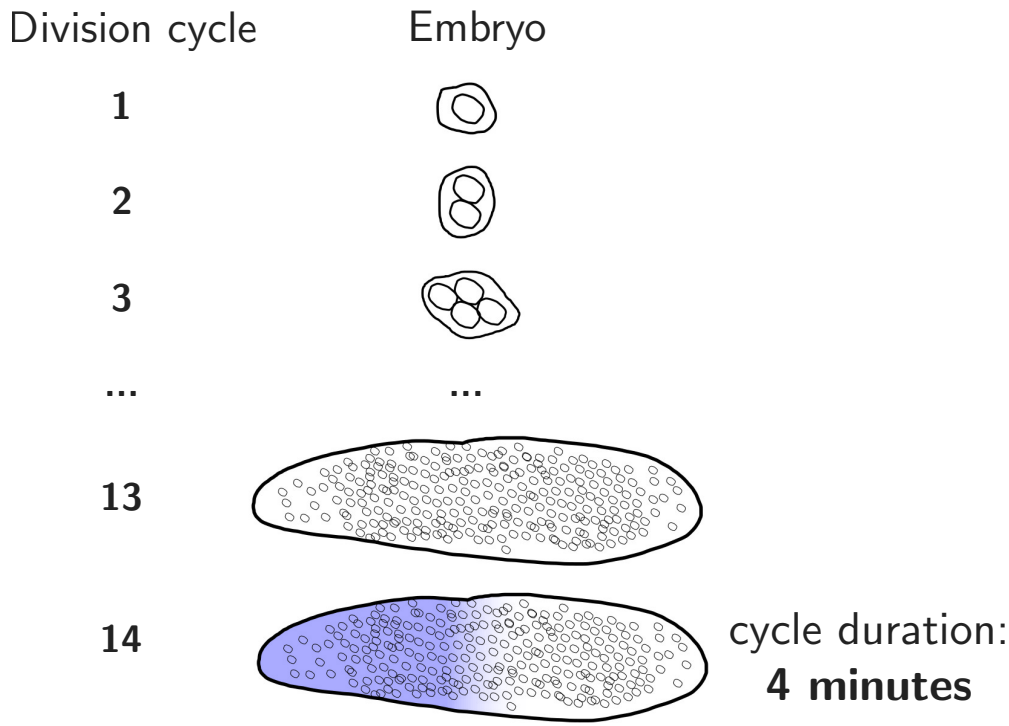
scene7

Precise sensing on a tight schedule



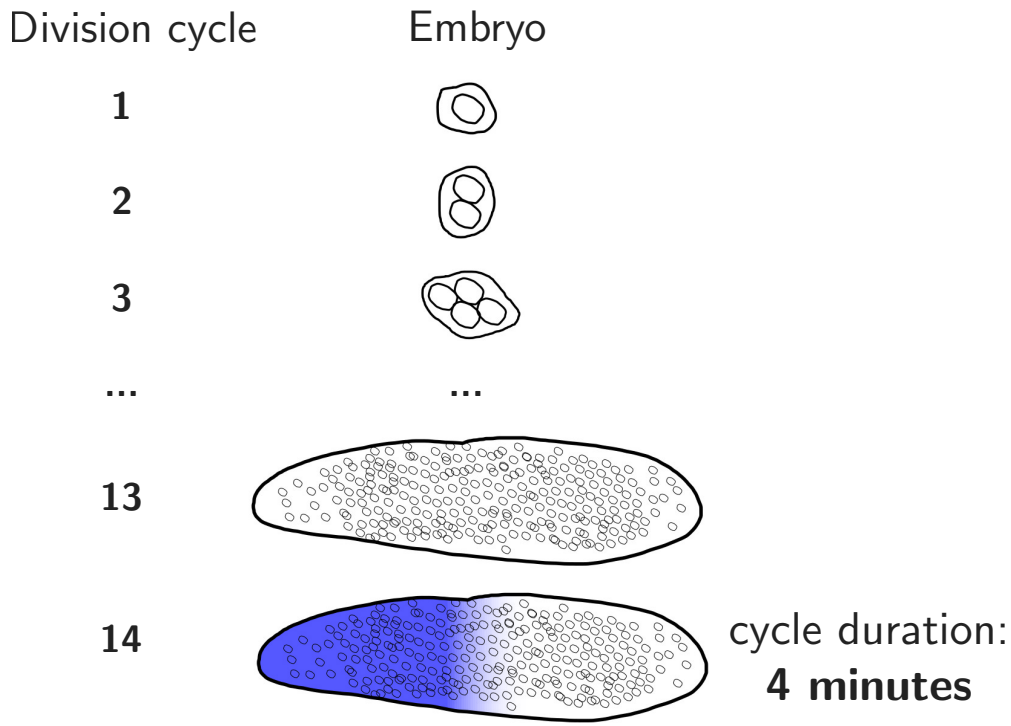
scene8_1

Precise sensing on a tight schedule



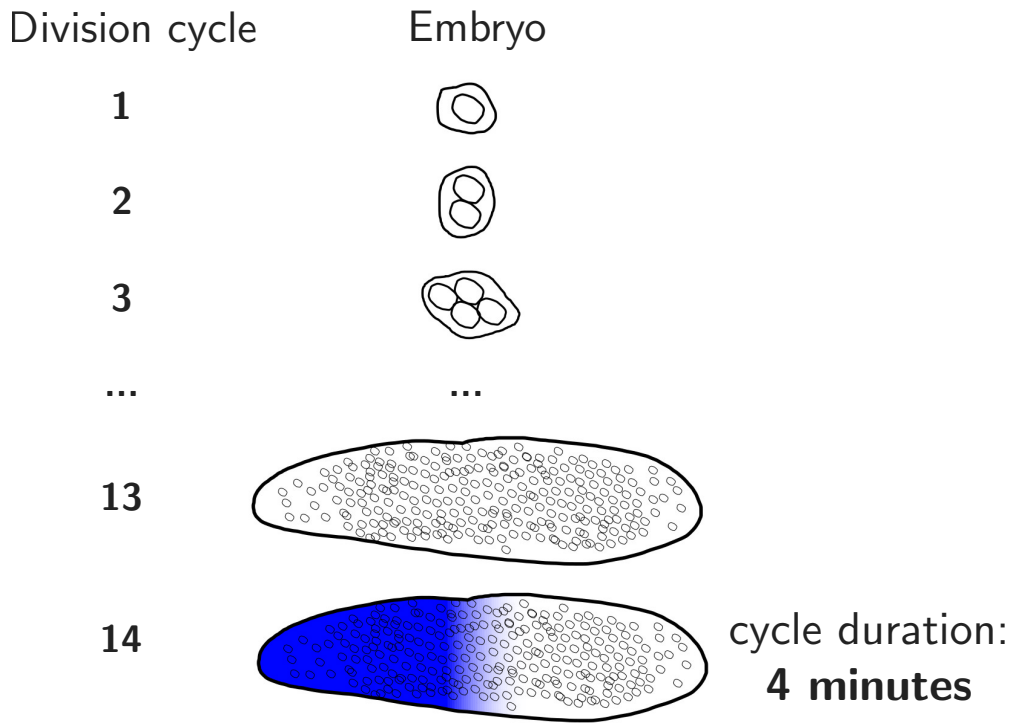
scene8_2

Precise sensing on a tight schedule



scene8_3

Precise sensing on a tight schedule



scene8_4

Understanding the precision of concentration sensing

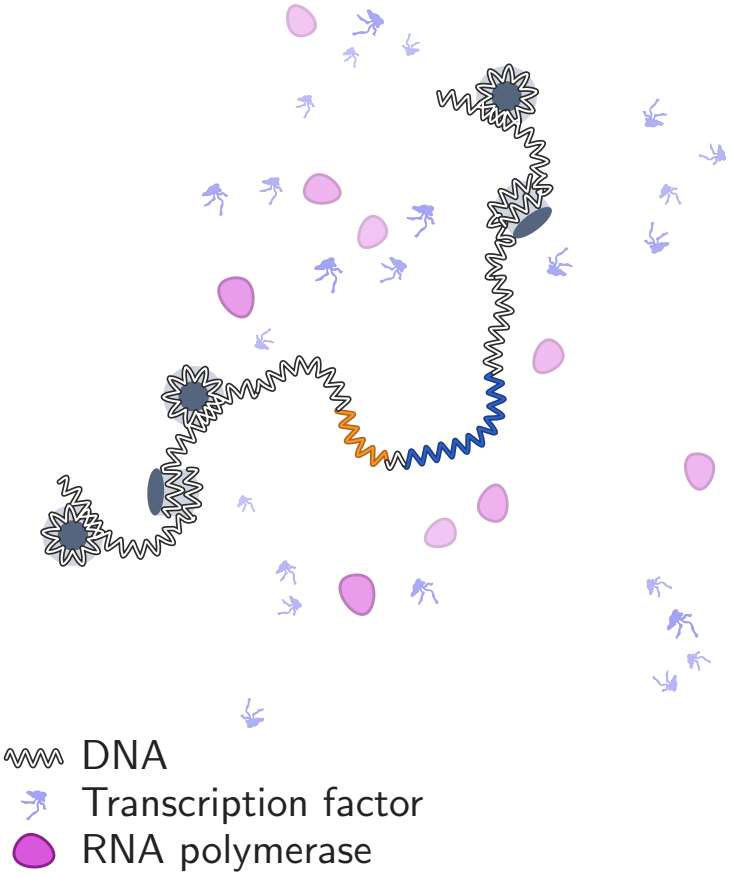
How to quantify precision?
Need a way to compare across models
Focusing on a promoter: optimal architectures
Polymer and condensation effects on precision

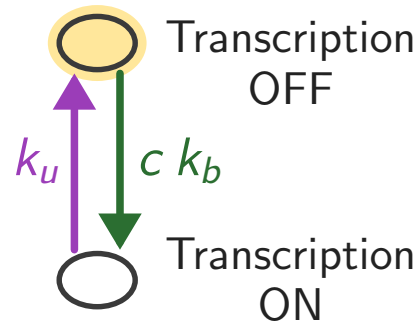
scene9

Understanding the precision of concentration sensing

scene10

A simple model of transcriptional control

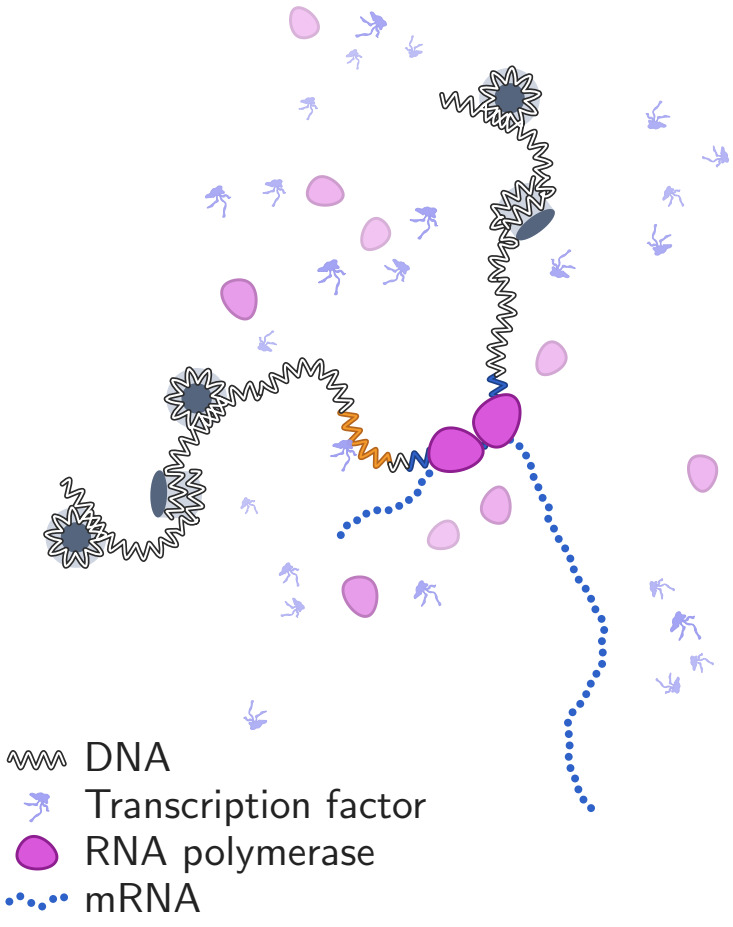


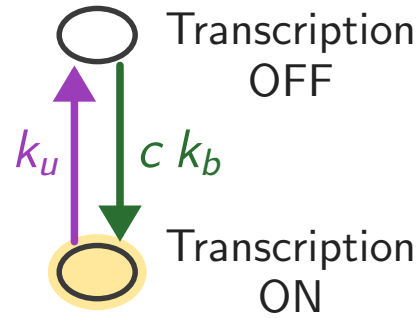




scene11_1

A simple model of transcriptional control

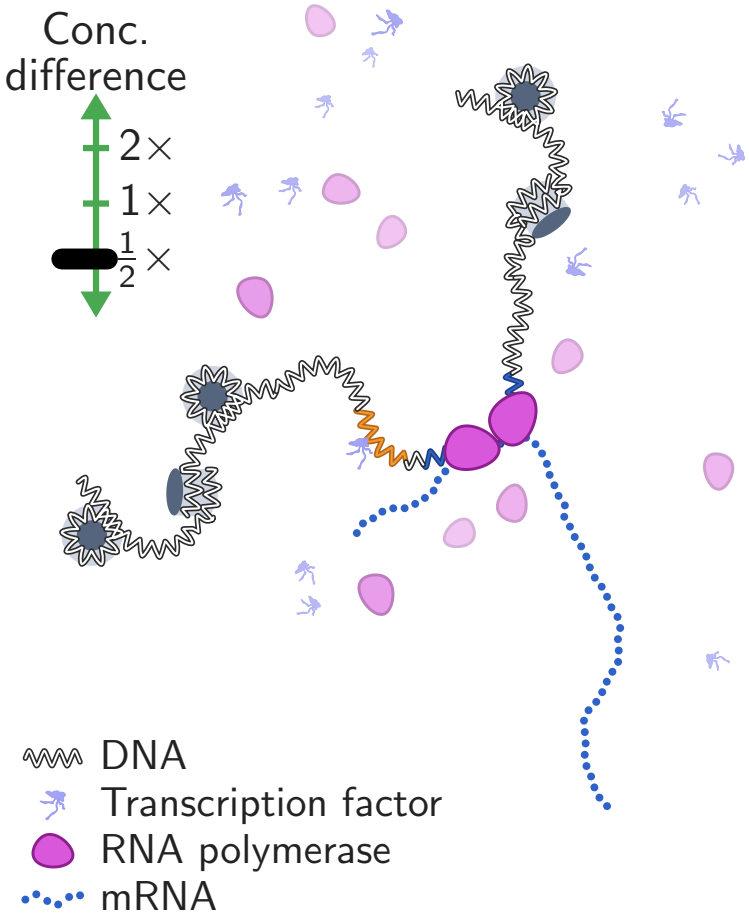


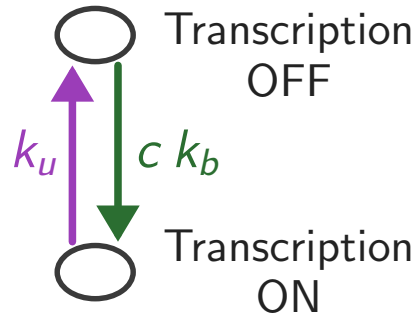


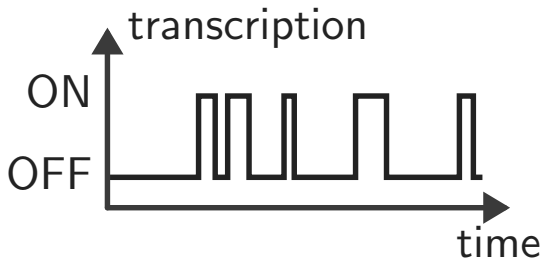


scene11_2

A simple model of transcriptional control

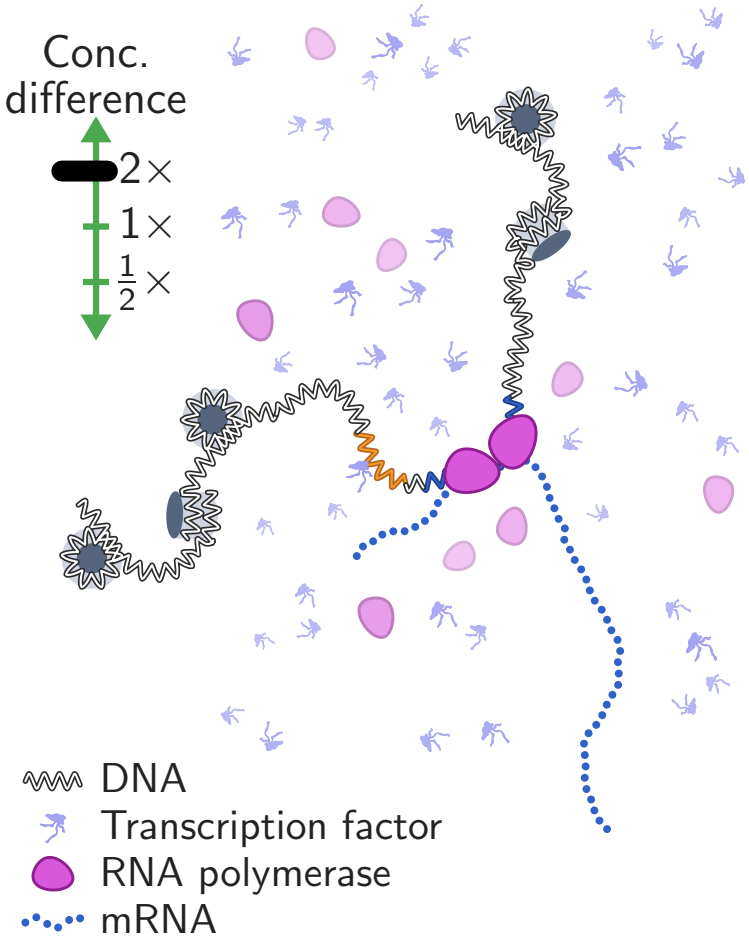


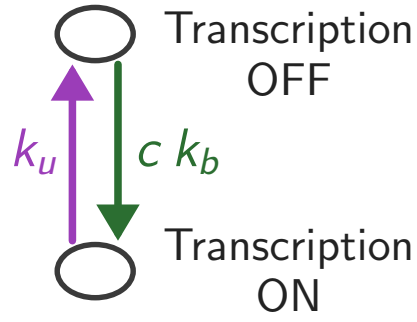


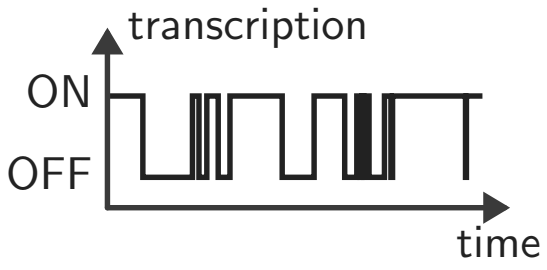


scene11_3

A simple model of transcriptional control







scene11_4

Sequential hypothesis testing

$$P(\text{} \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

$$\frac{P(\text{  } \mid c_{\text{low}})}{P(\text{  } \mid c_{\text{high}})} > 1$$

scene12_1

Sequential hypothesis testing

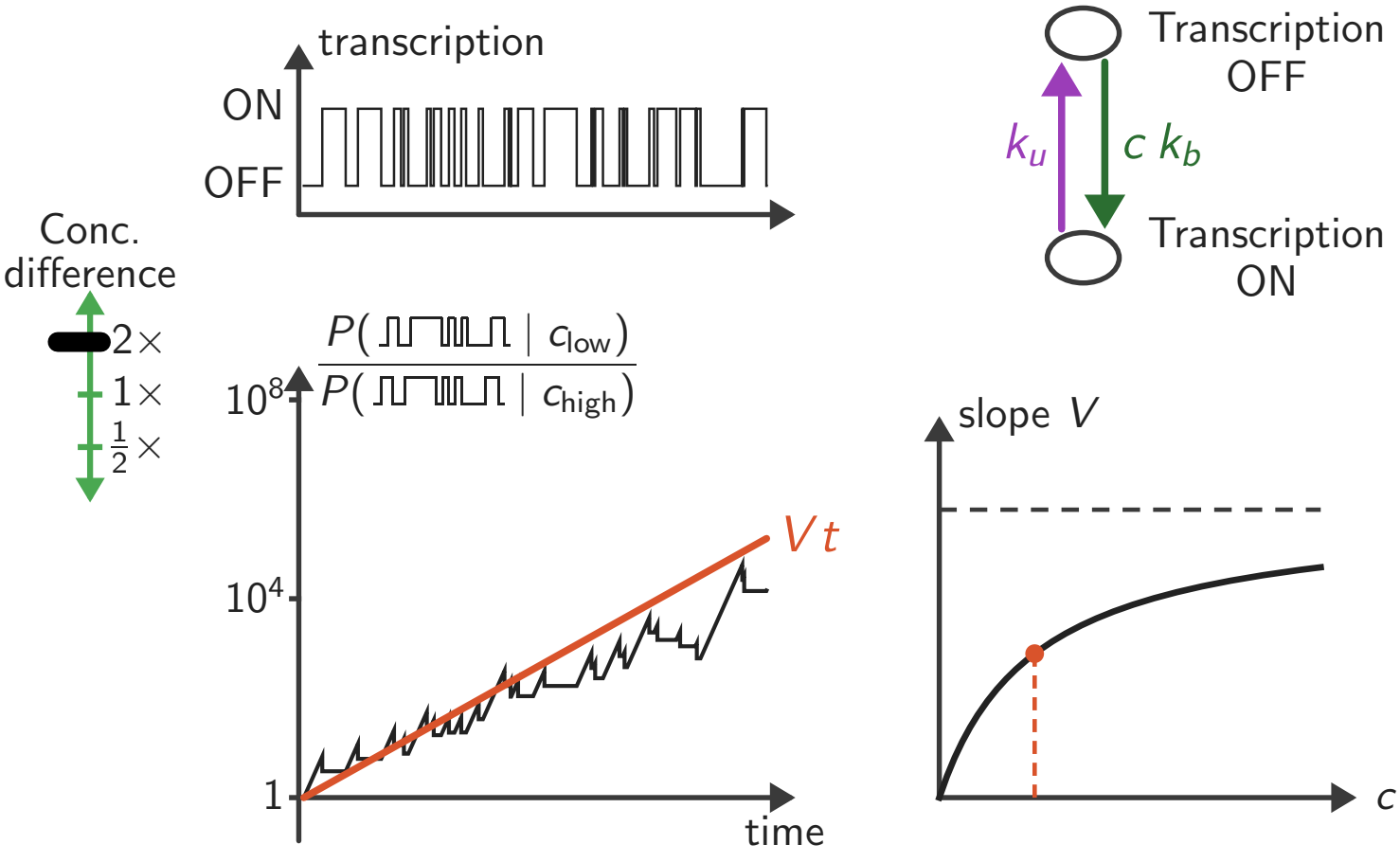
$$P(\text{} \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

$$R(t) = \frac{P(\text{  } \mid c_{\text{low}})}{P(\text{} \mid c_{\text{high}})} > 1$$

$$\log R(t) = V t + \sqrt{2D}\,\xi$$

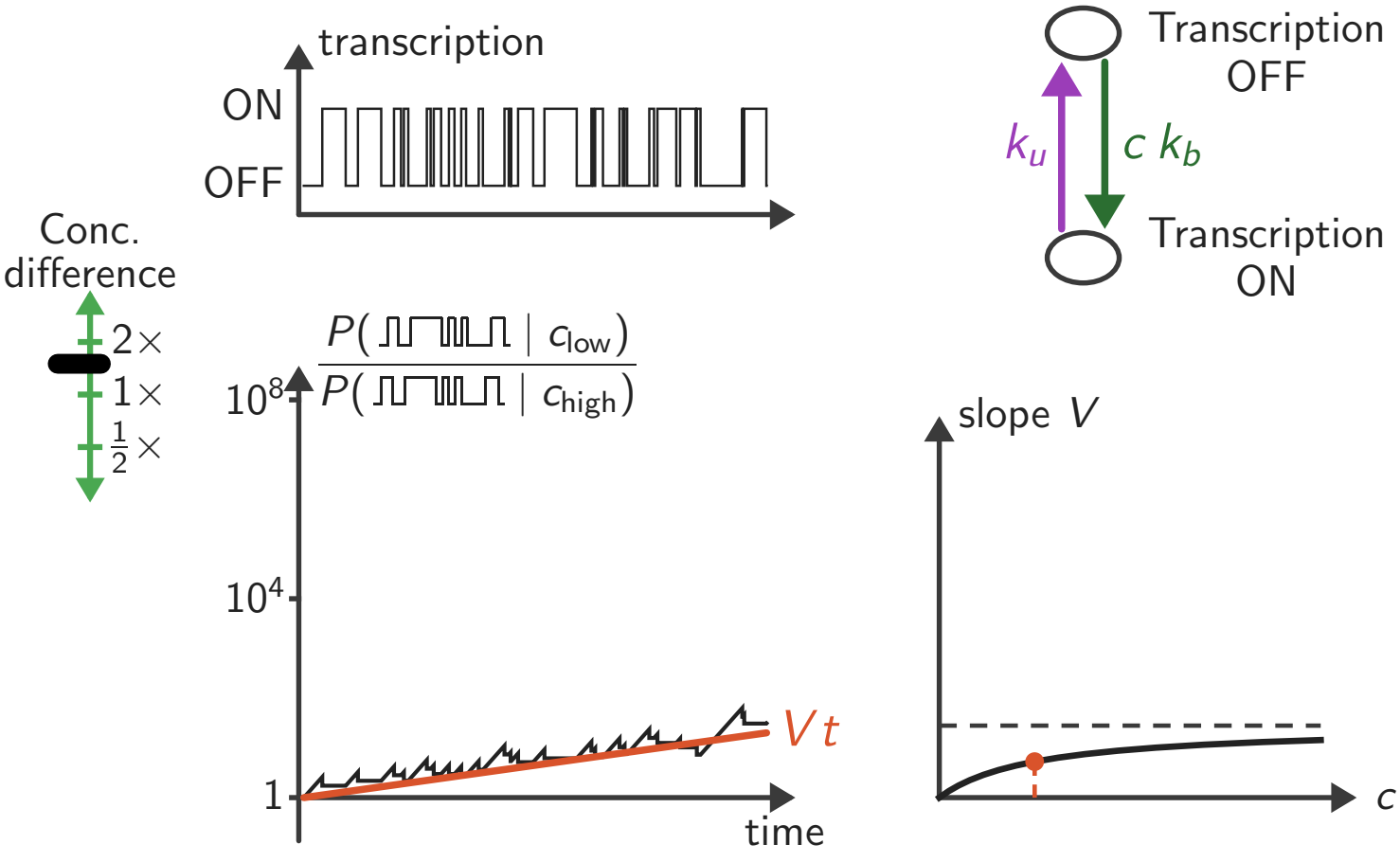
scene12_2

Sequential hypothesis testing



scene13_1

Sequential hypothesis testing



scene13_2

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

scene14_1

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

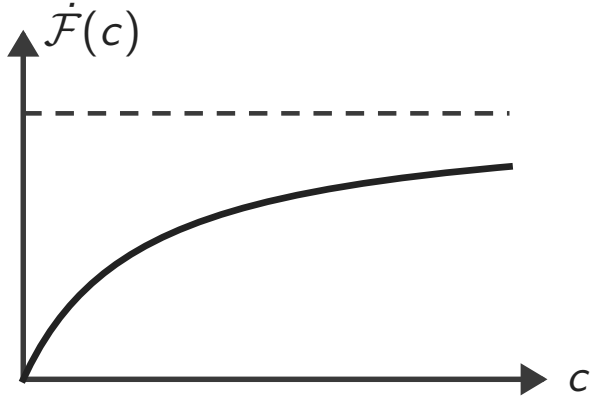
scene14_2

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$



$c\,k_b$: diffusion-limited arrival of molecules

$k_b = 4aD$

scene14_3

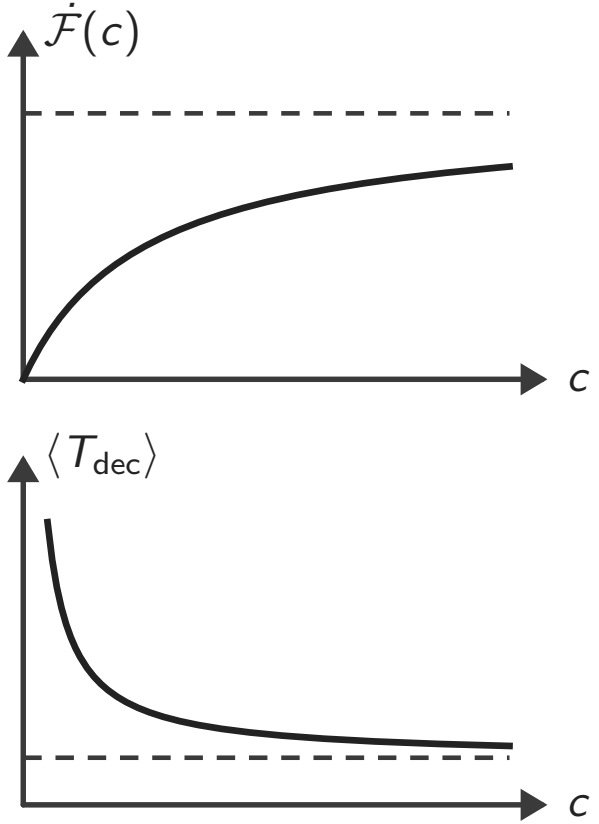
Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty}\frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2}\right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$

$$\langle T_{\text{dec}}\rangle = \frac{K}{V}\tanh\left(\frac{K}{2}\right)$$

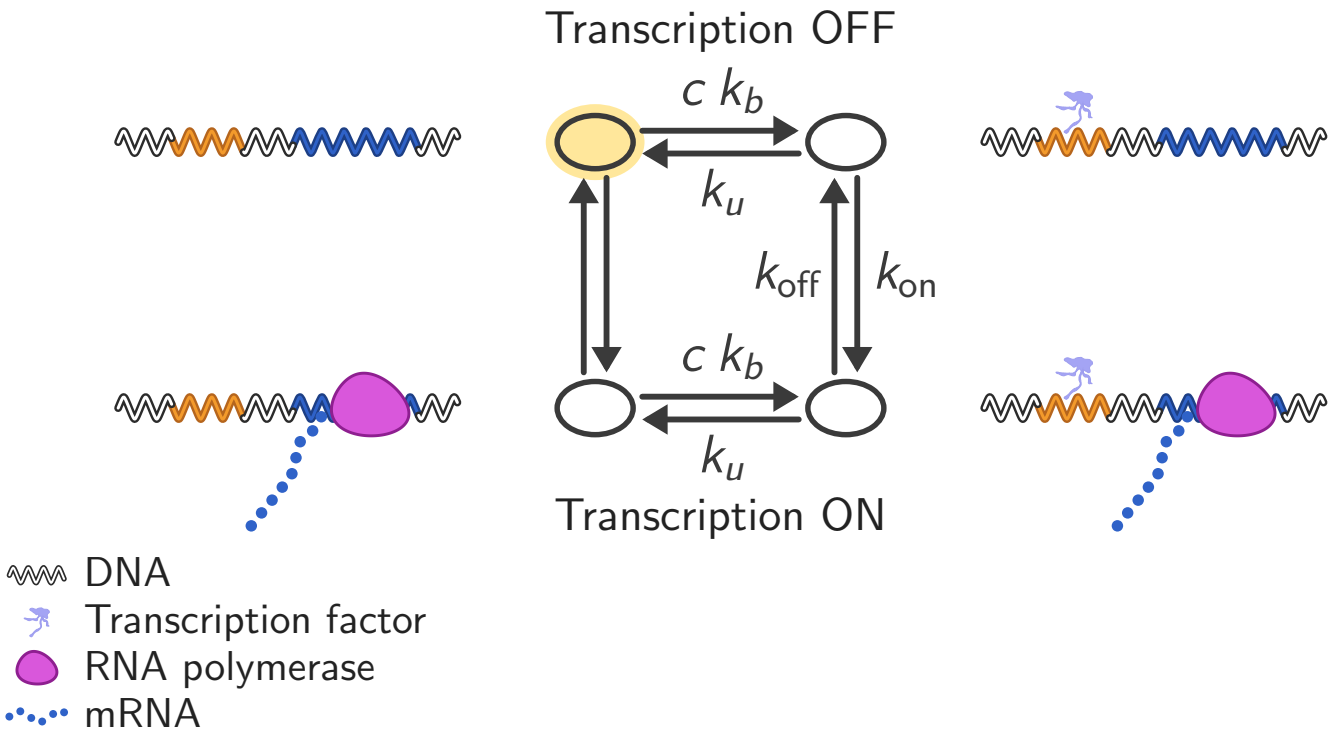


$c\,k_b$: diffusion-limited arrival of molecules

$k_b = 4aD$

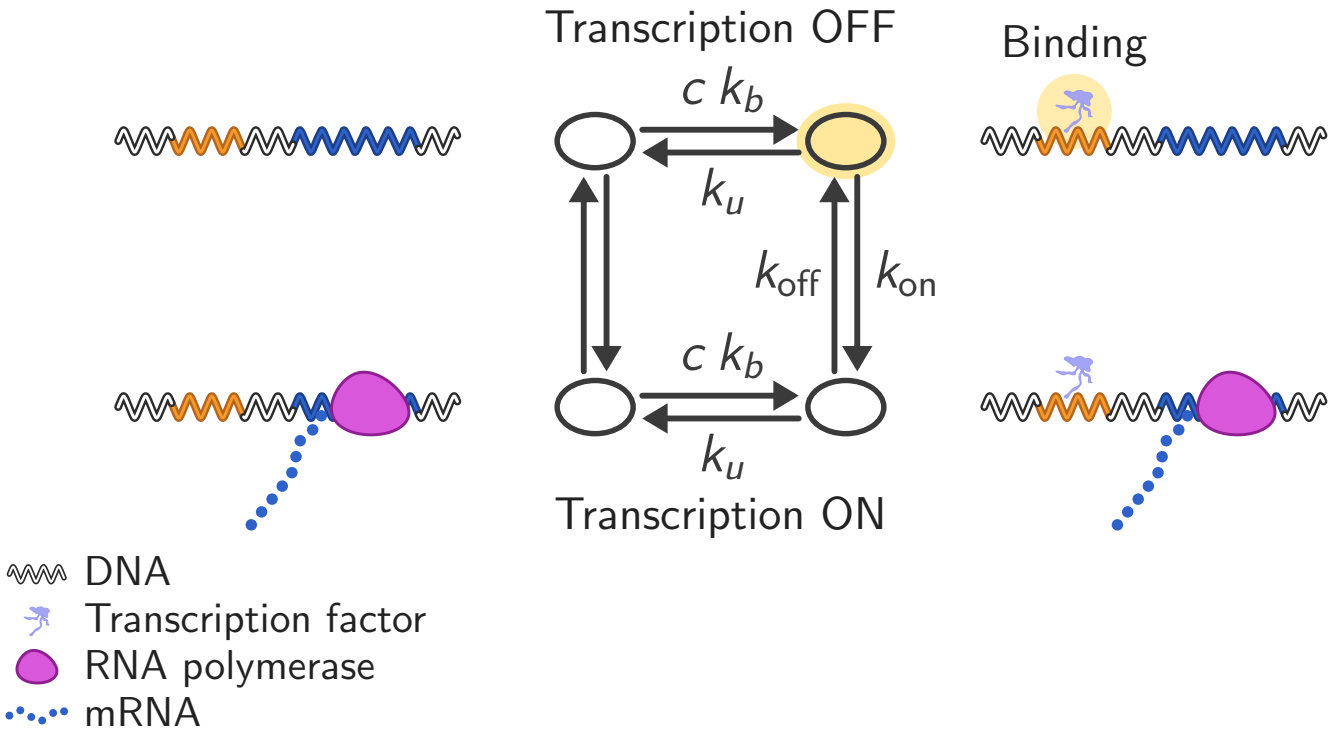
scene14$_4$

Modeling time-limited activation



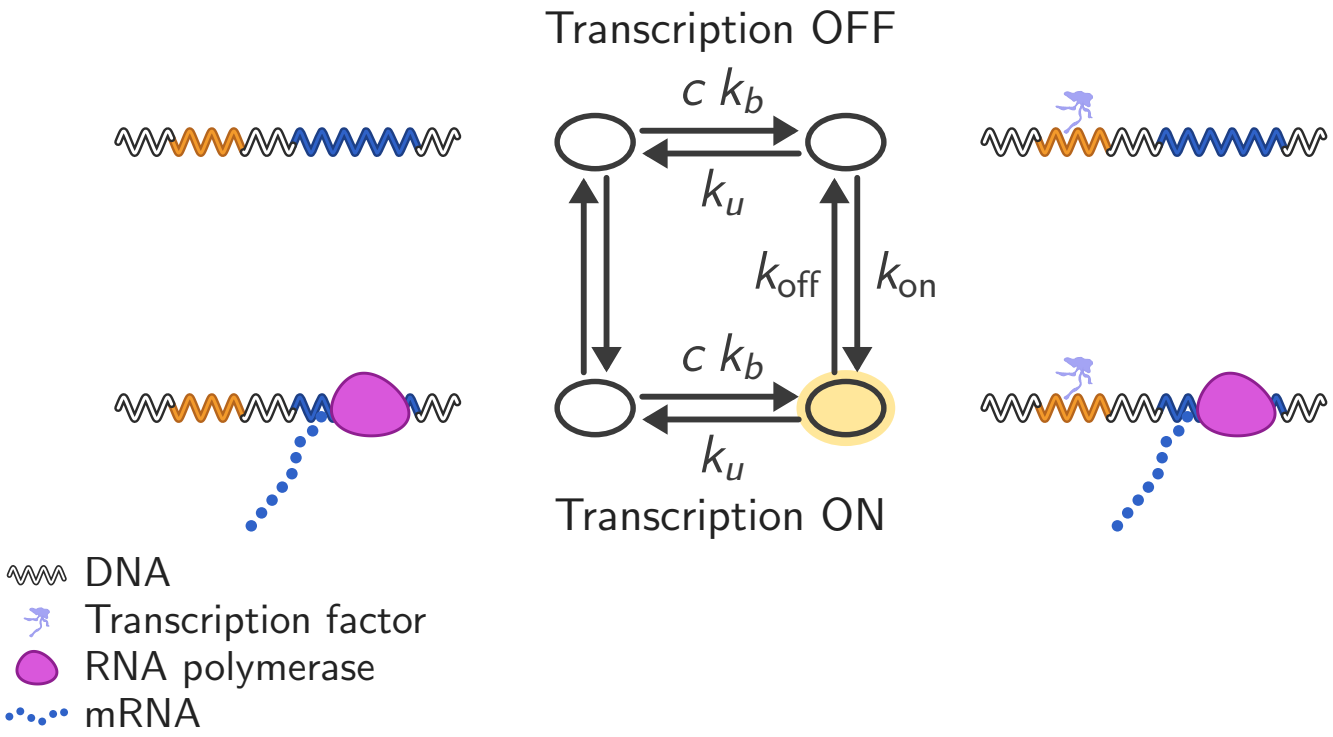
scene15_1

Modeling time-limited activation



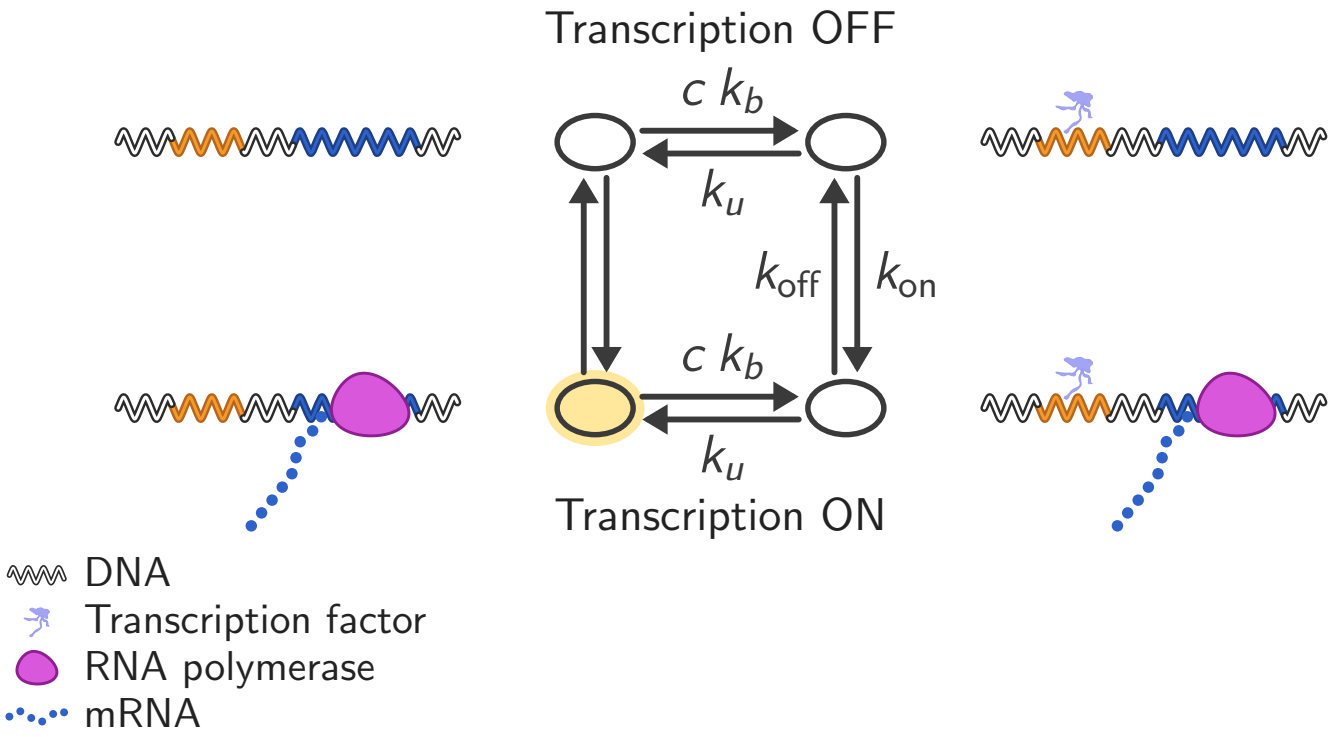
scene15_2

Modeling time-limited activation



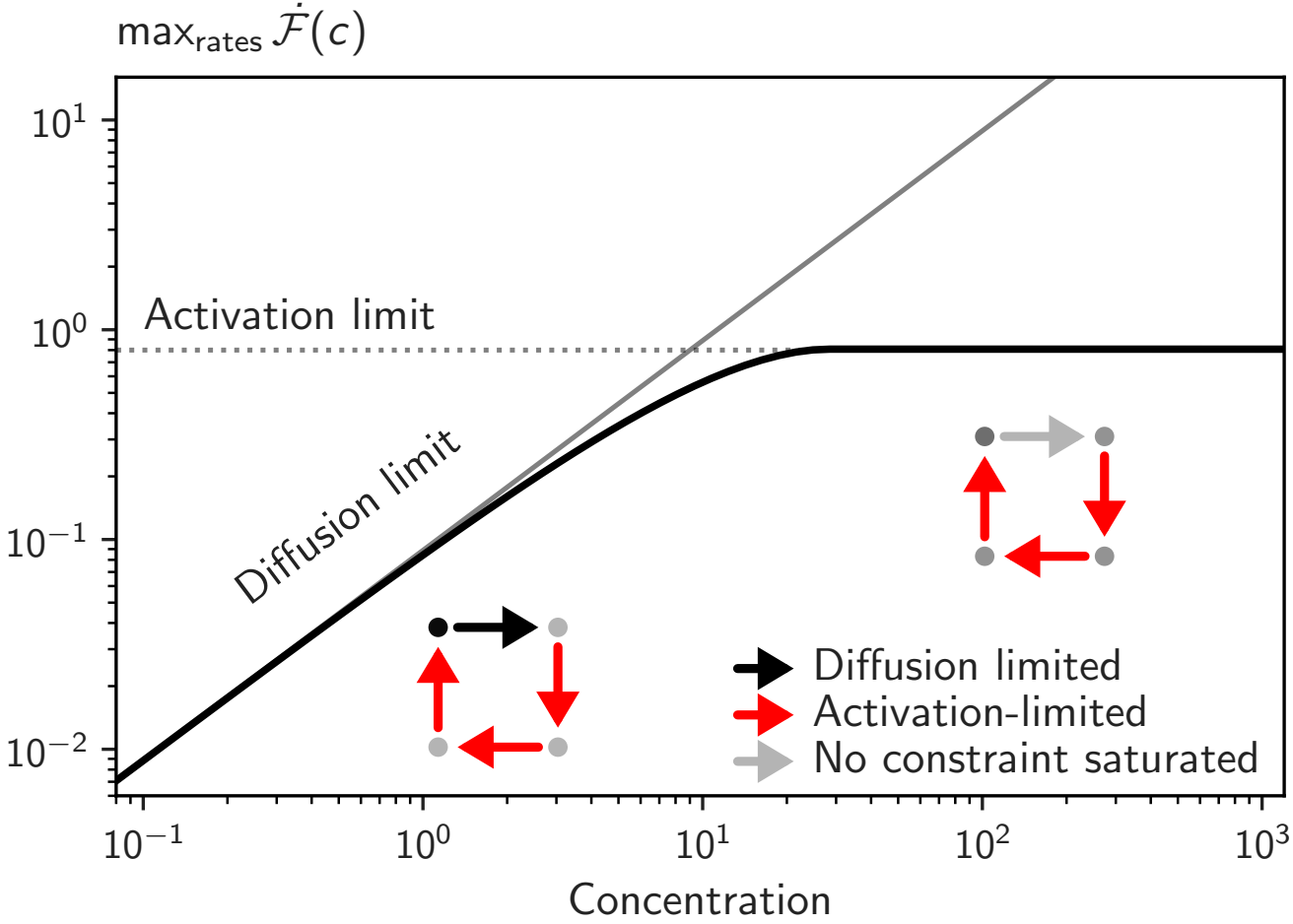
scene15_3

Modeling time-limited activation



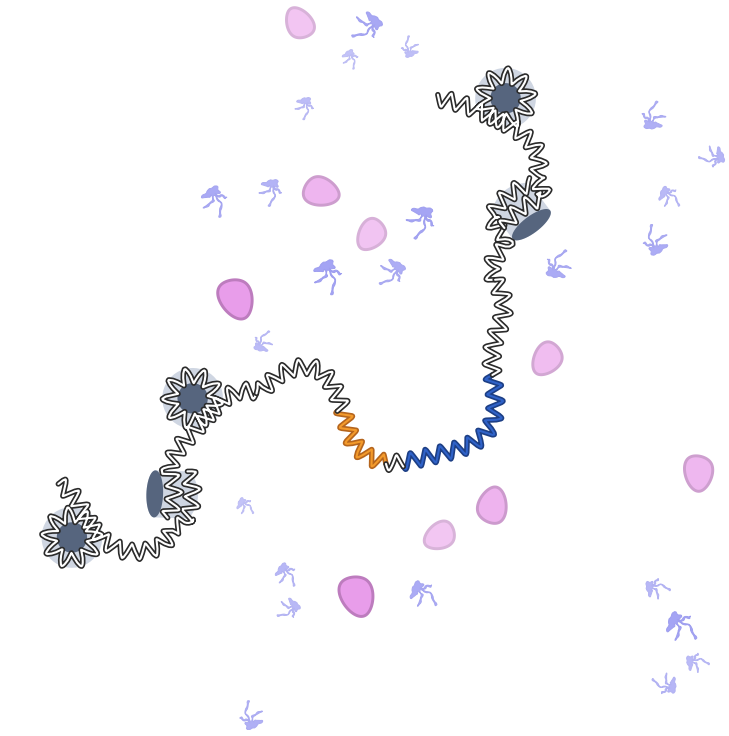
scene15_4

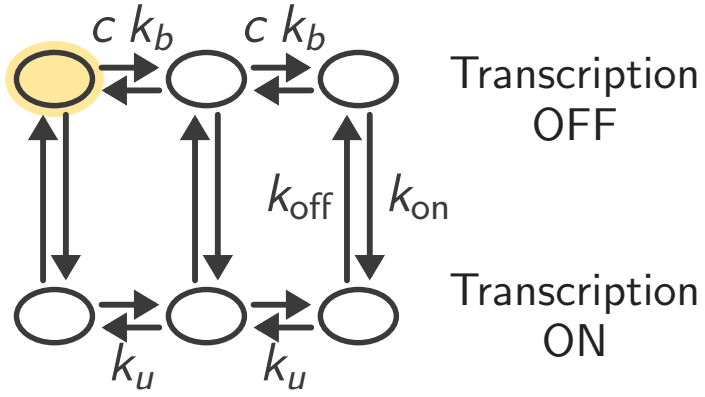
Optimal promoters: two distinct regimes



scene16

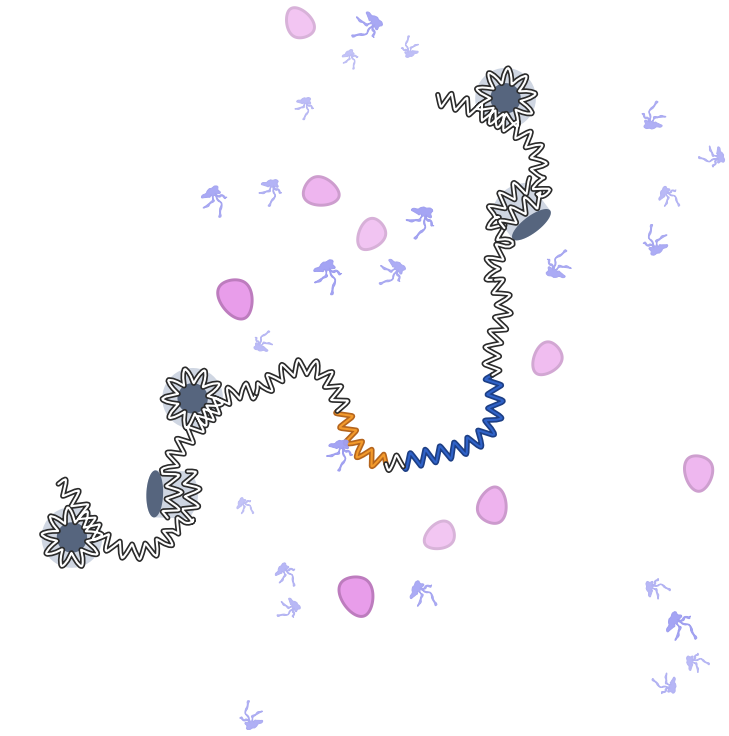
Cooperative binding

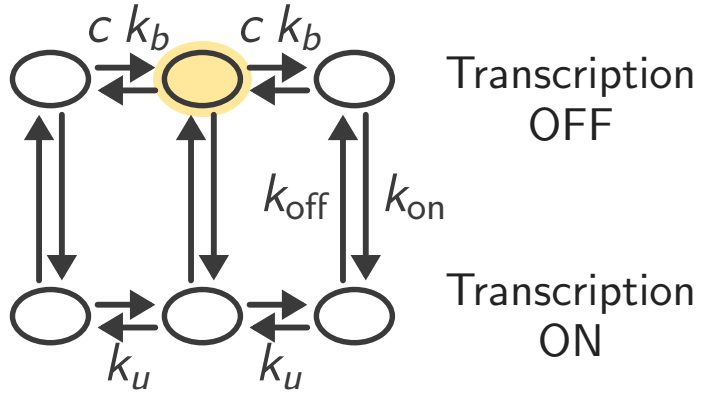




scene17_1

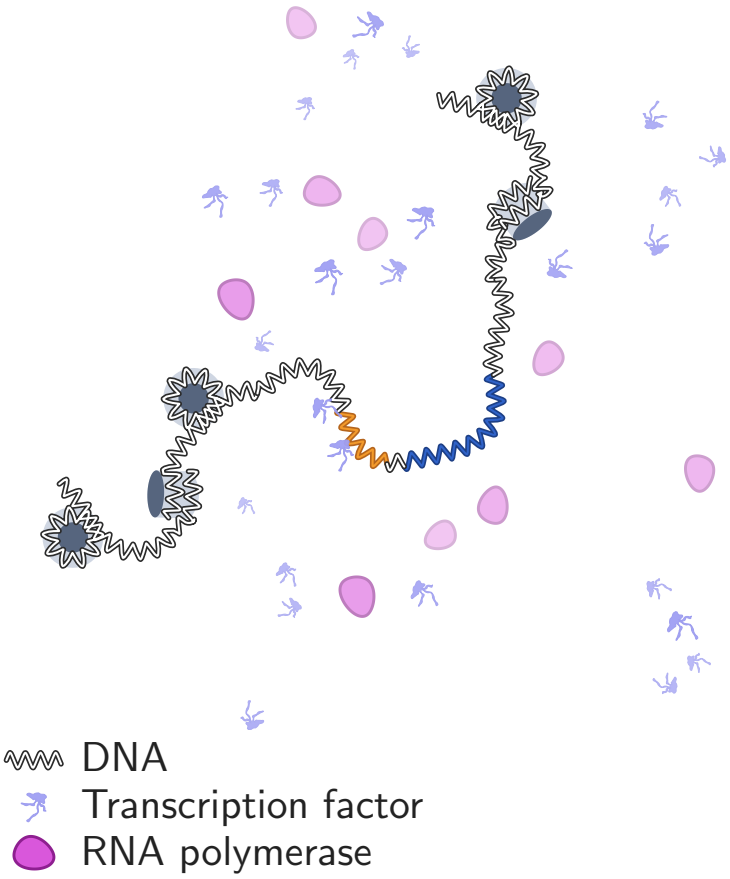
Cooperative binding

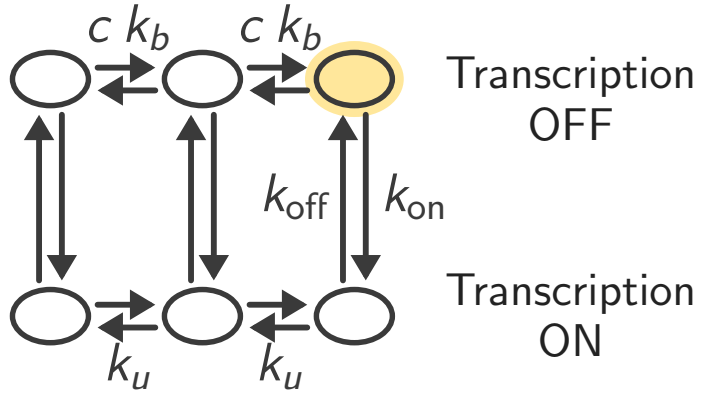




scene17_2

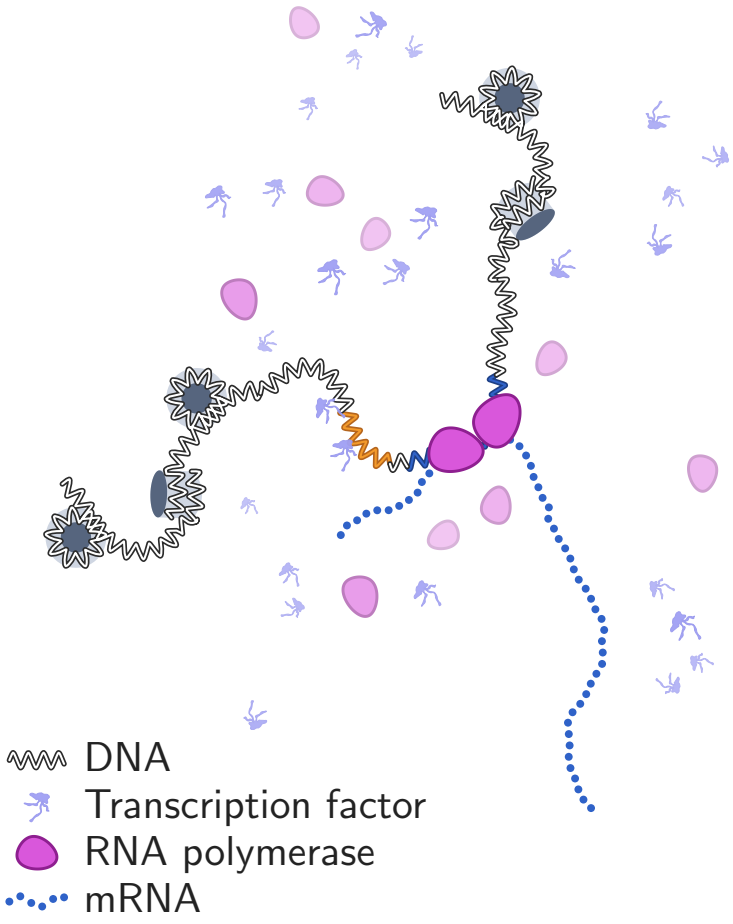
Cooperative binding

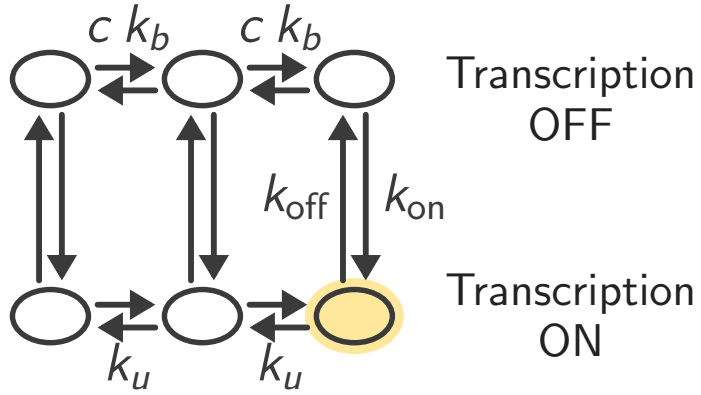




scene17_3

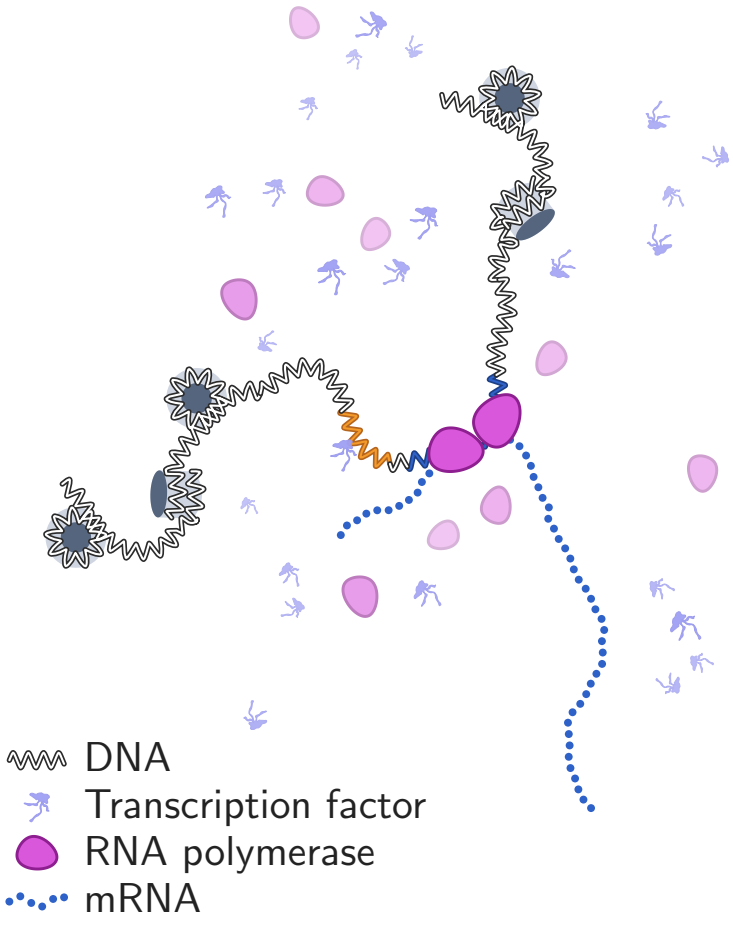
Cooperative binding

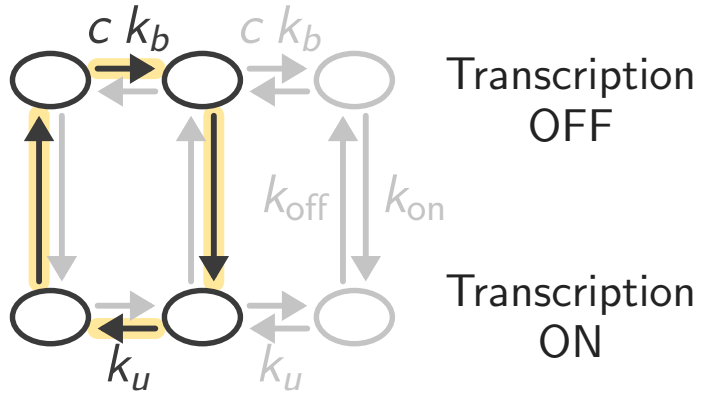


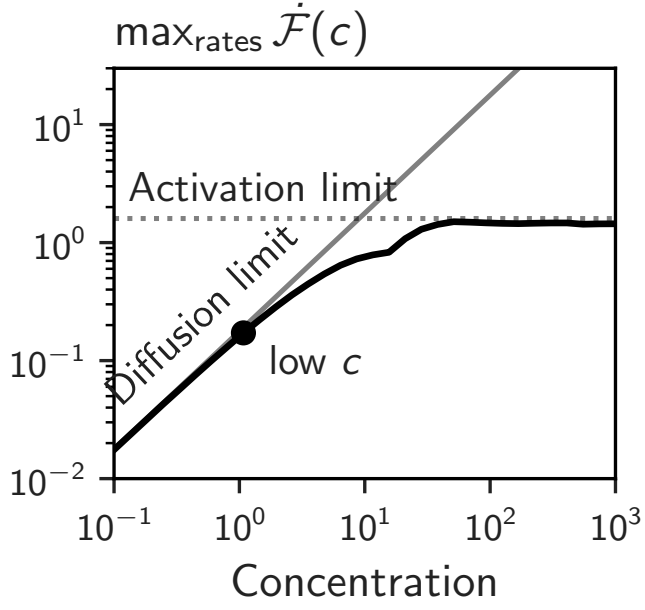


scene17_4

Cooperative binding

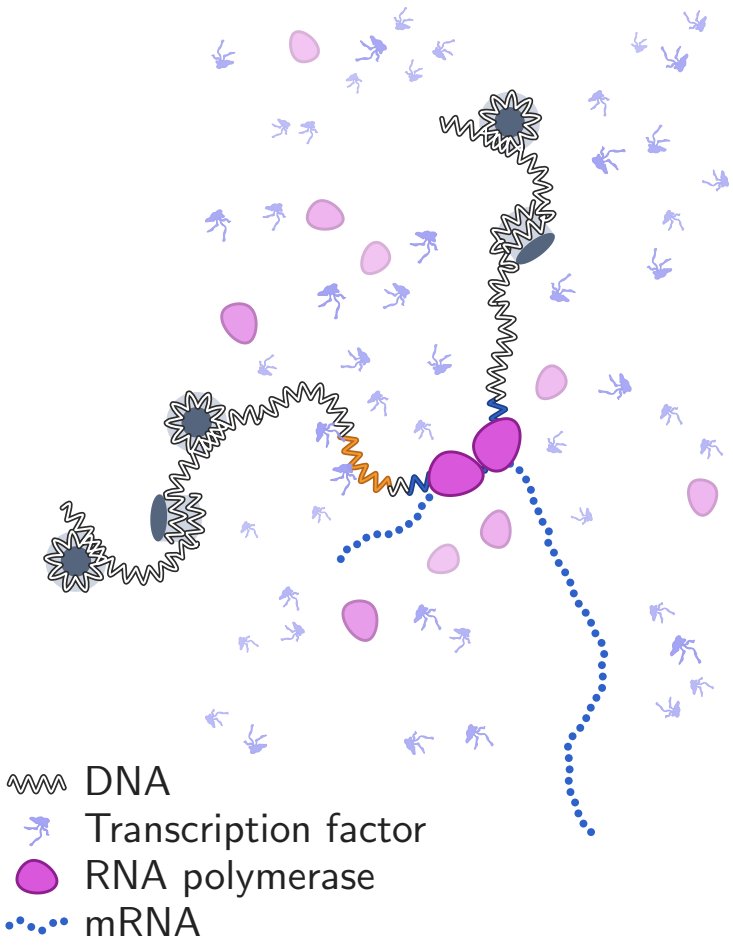


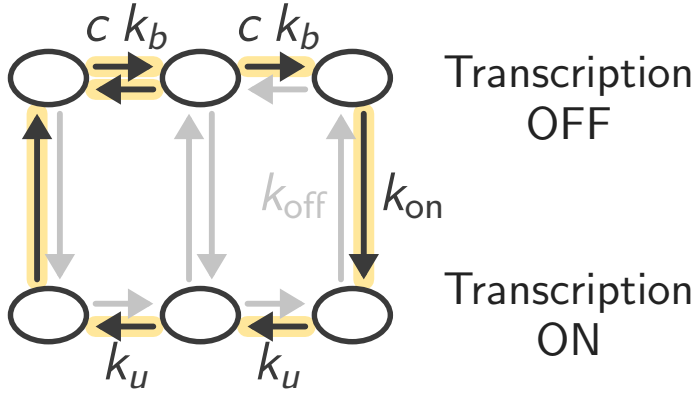


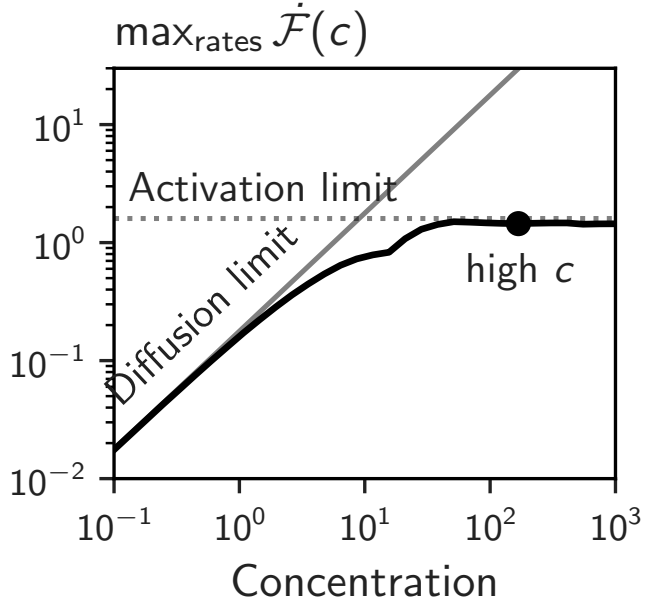


scene17_5

Cooperative binding

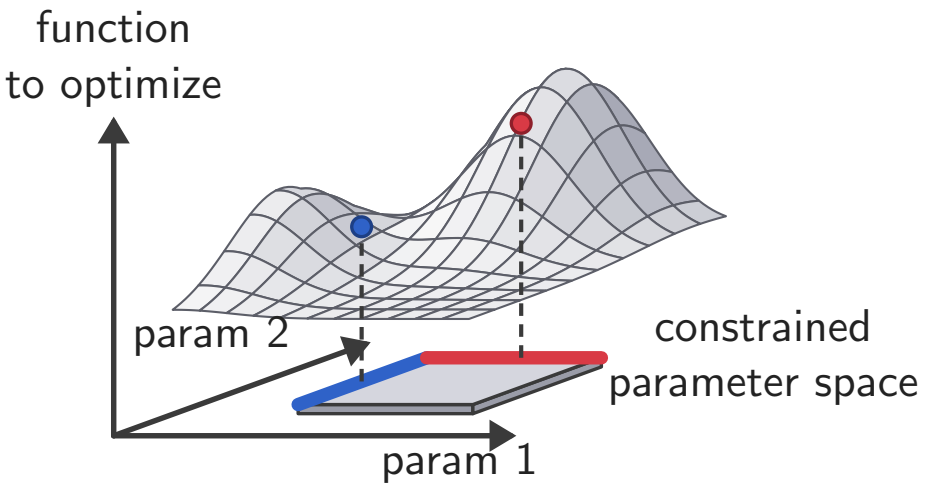




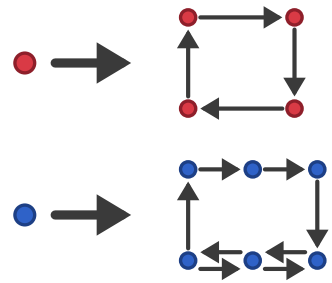


scene17$_6$

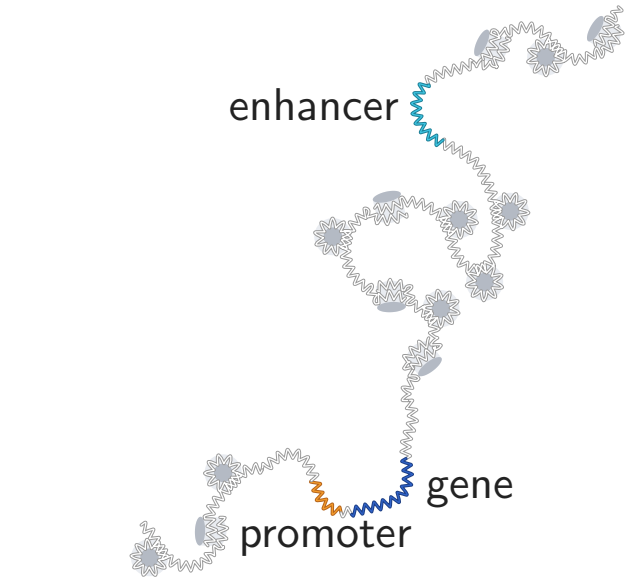
Takeaway



same optimized function
different mechanisms



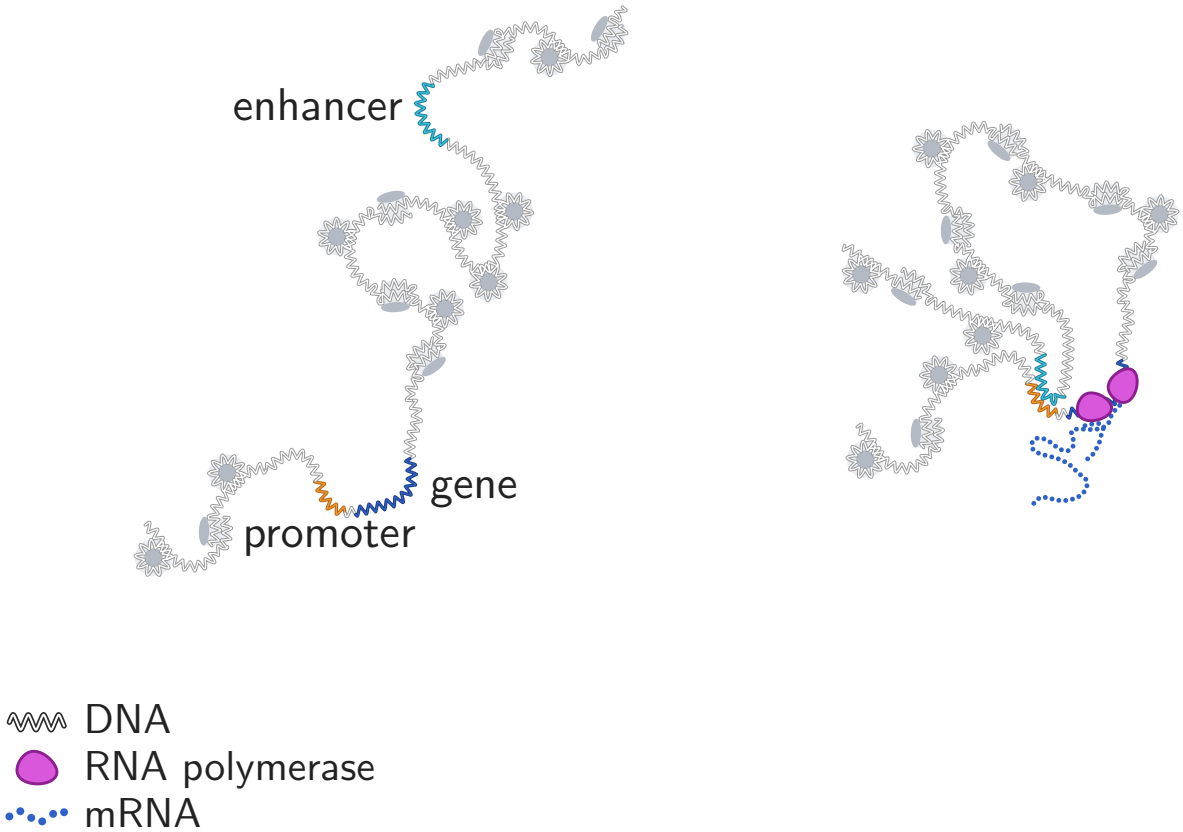
Transcriptional control by distal elements



DNA

scene20_1

Transcriptional control by distal elements

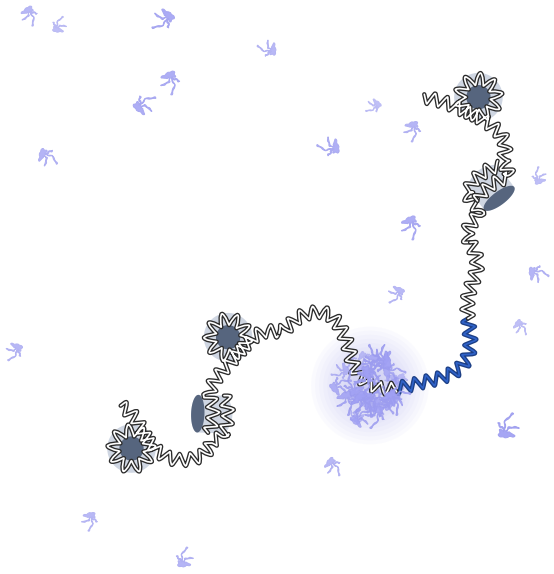


scene20_2

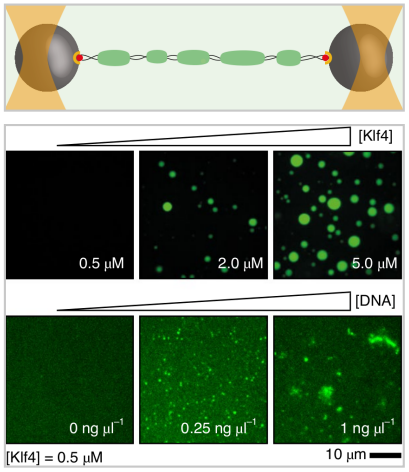
Transcriptional control through liquid phase separation?

scene21$_1$

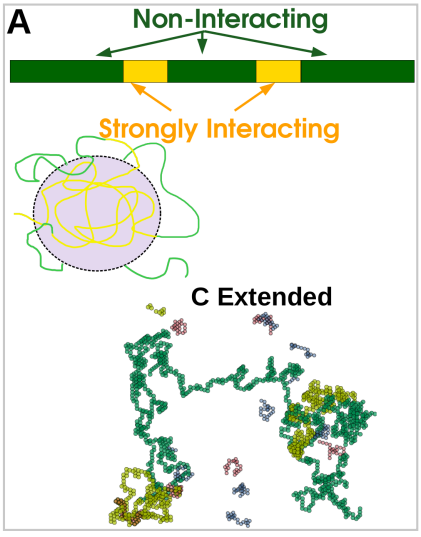
Proteins condense on the surface of DNA



DNA
Transcription factor



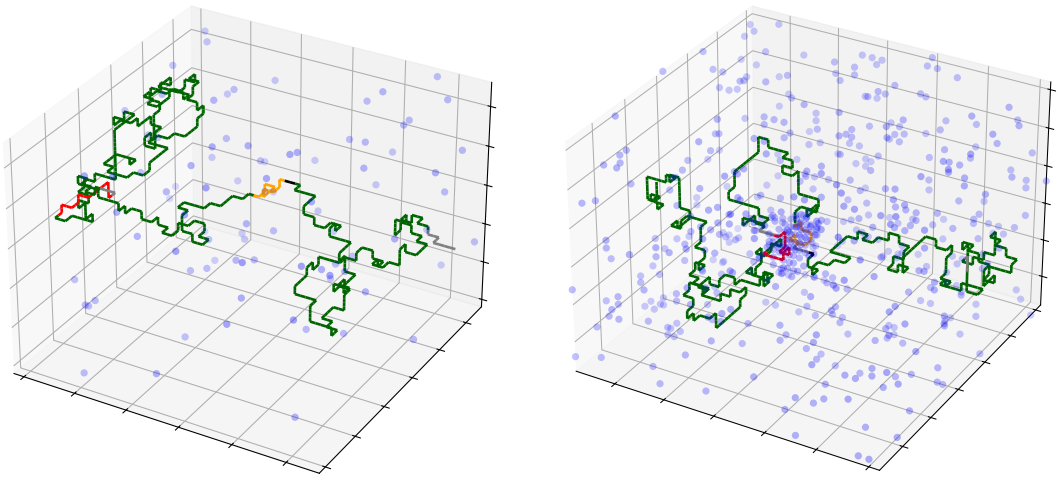
Sequence-dependent surface condensation of a pioneer transcription factor on DNA
Morin et al., 2022

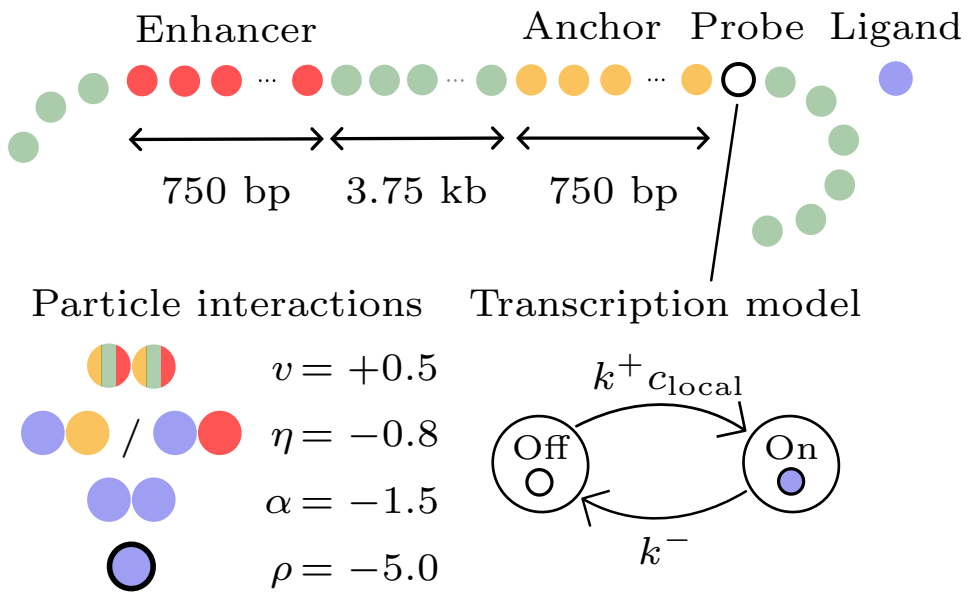


Protein-DNA Co-condensation is Prewetting to a Collapsed Polymer
Rouches & Machta, 2025

scene21$_2$

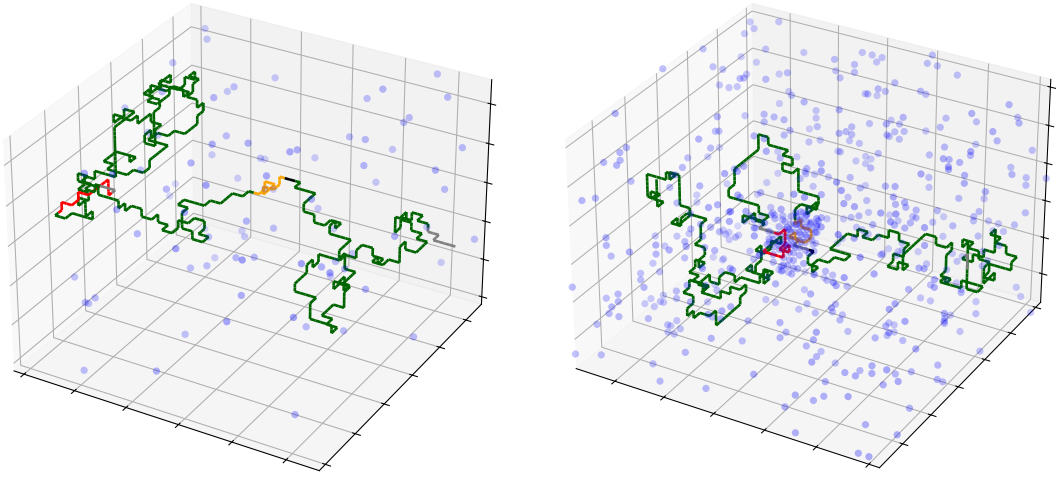
Coupling condensation and polymer dynamics

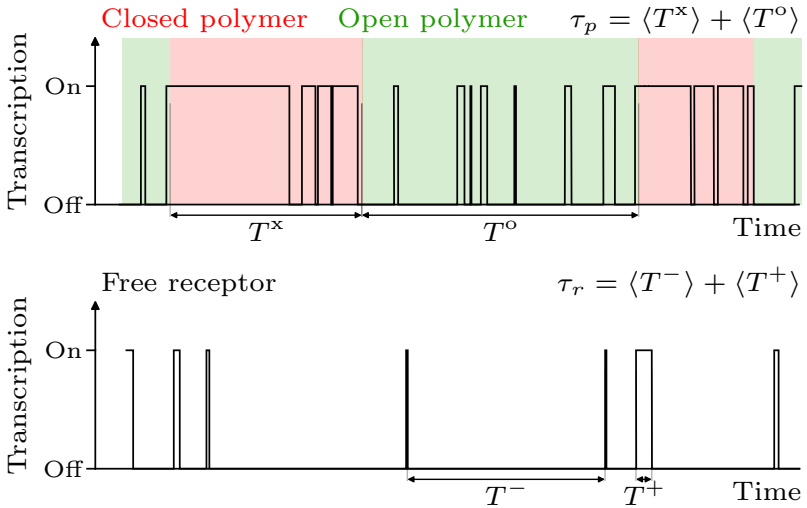


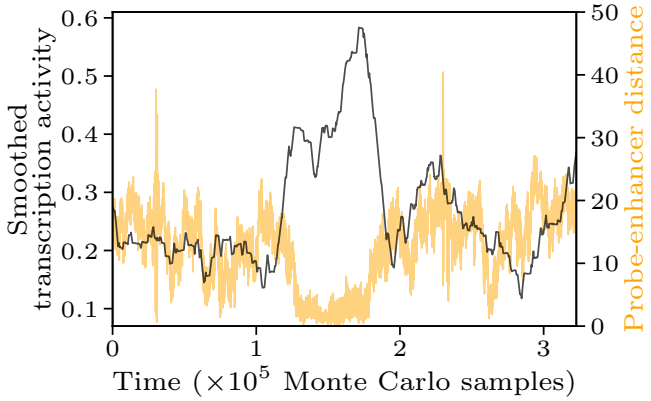


scene22

Coupling condensation and polymer dynamics

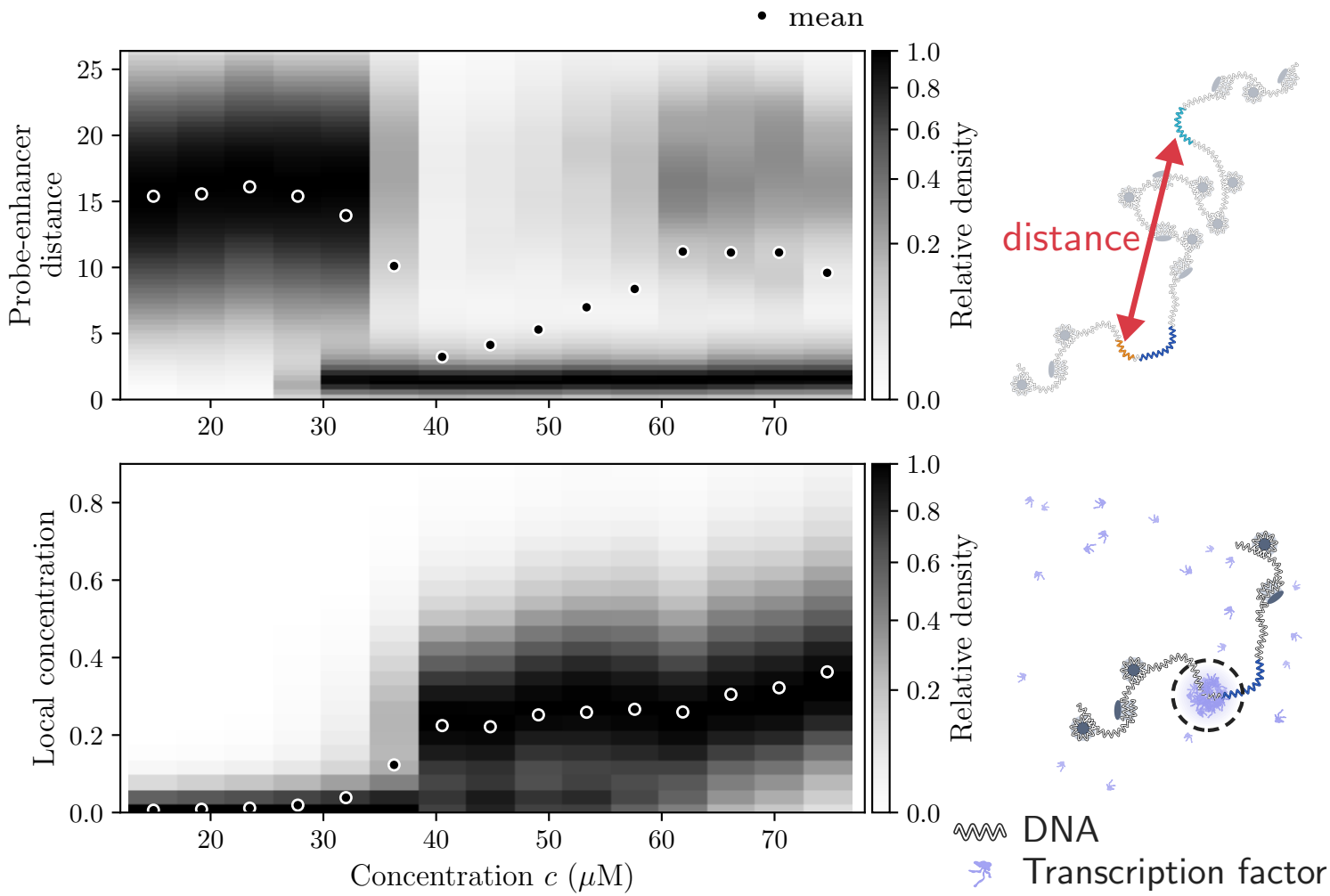




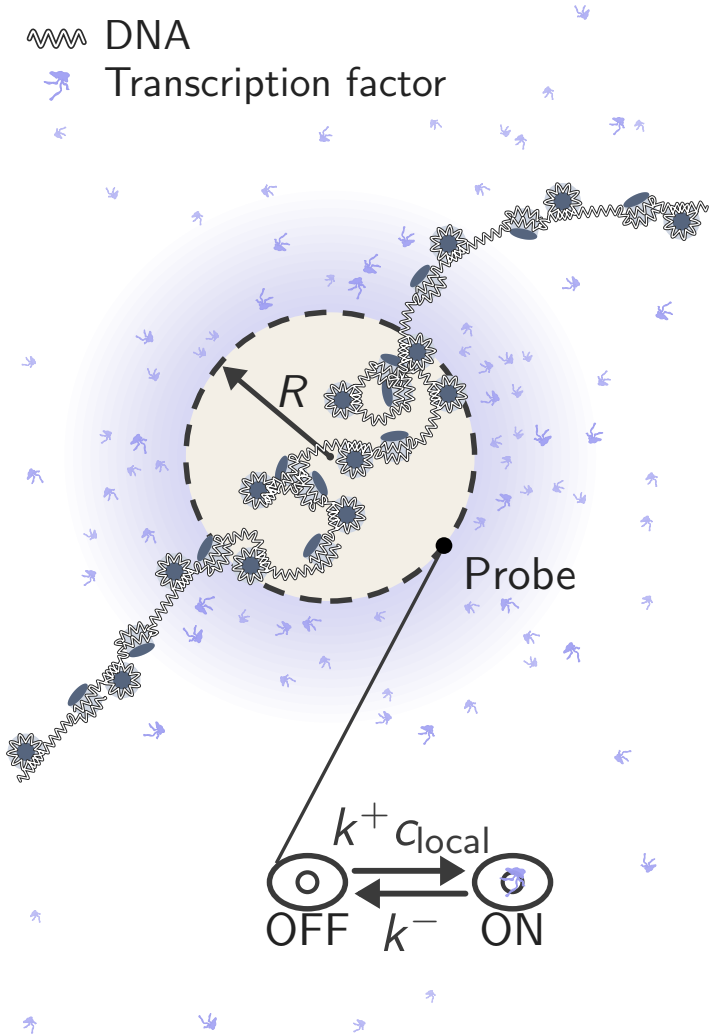


scene23$_2$

Polymer configuration depends sensitively on concentration



scene24



Flory–Huggins potential

$$f(\phi) = \frac{\phi}{\nu} \ln \phi + (1-\phi) \ln(1-\phi) - \chi\phi^2 - \mu\phi$$

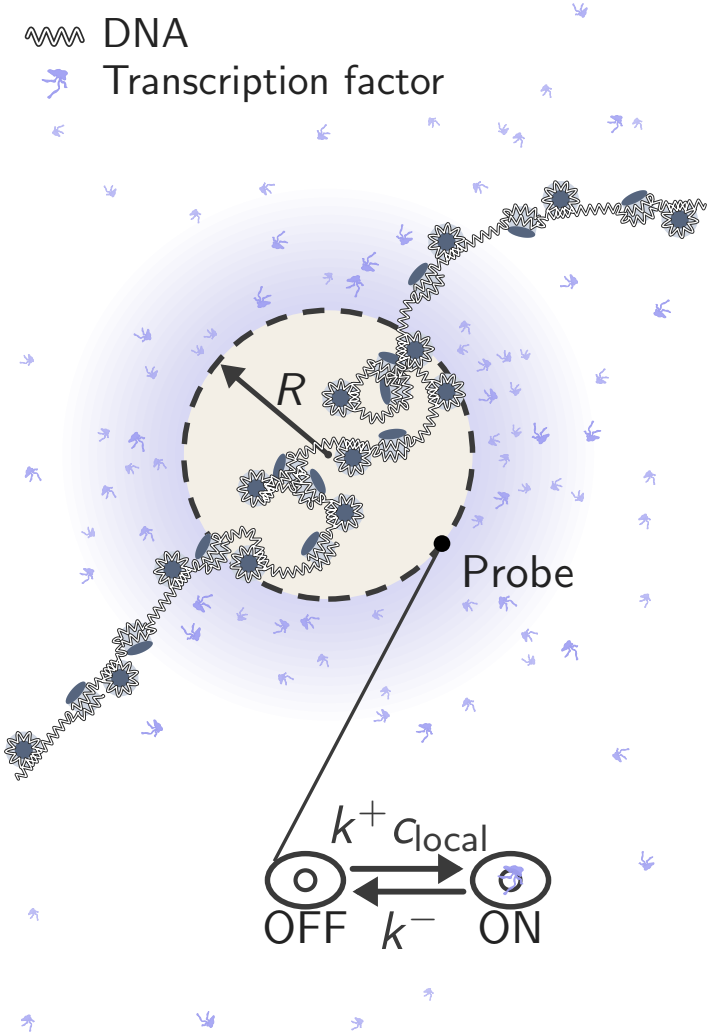
Cahn–Hilliard equation

$$0 = \nabla^2 \left[f'(\phi) - \kappa \nabla^2 \phi \right]$$

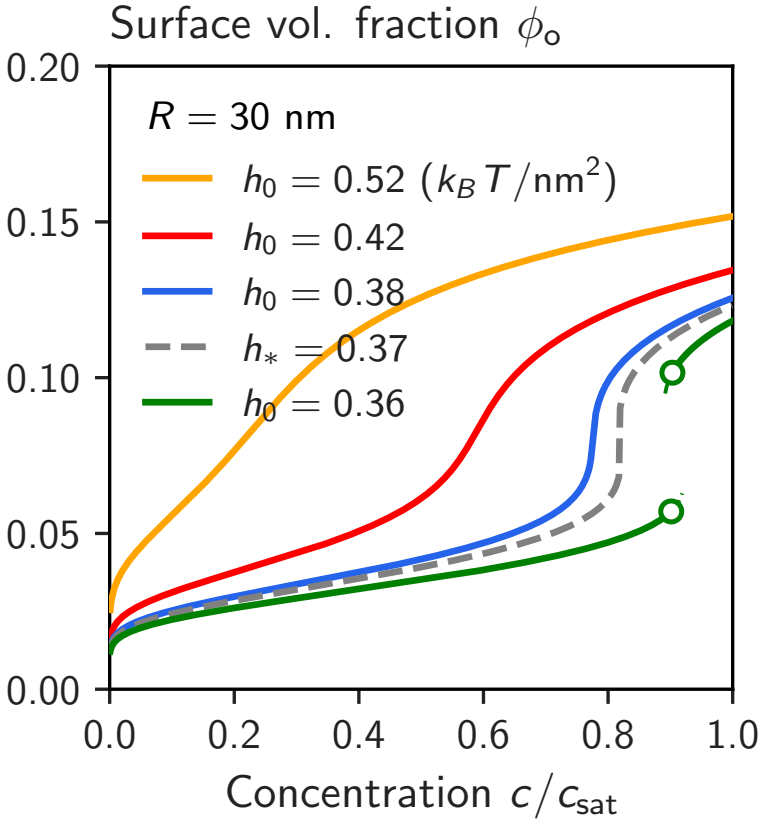
$$\kappa \, \partial_r \phi \big|_{r=R} = -h_0$$

ϕ	TF volume fraction
ν	TF / solvent volume ratio
χ	TF–TF attraction
μ	chemical potential
h_0	surface attraction
κ	interface stiffness

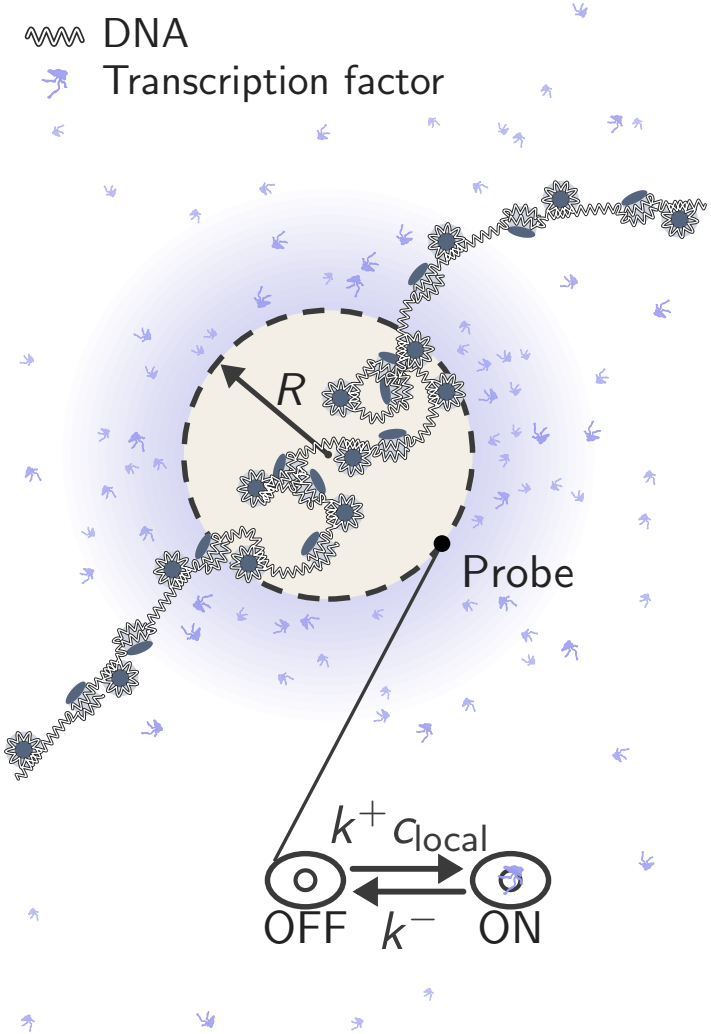
scene25$_1$



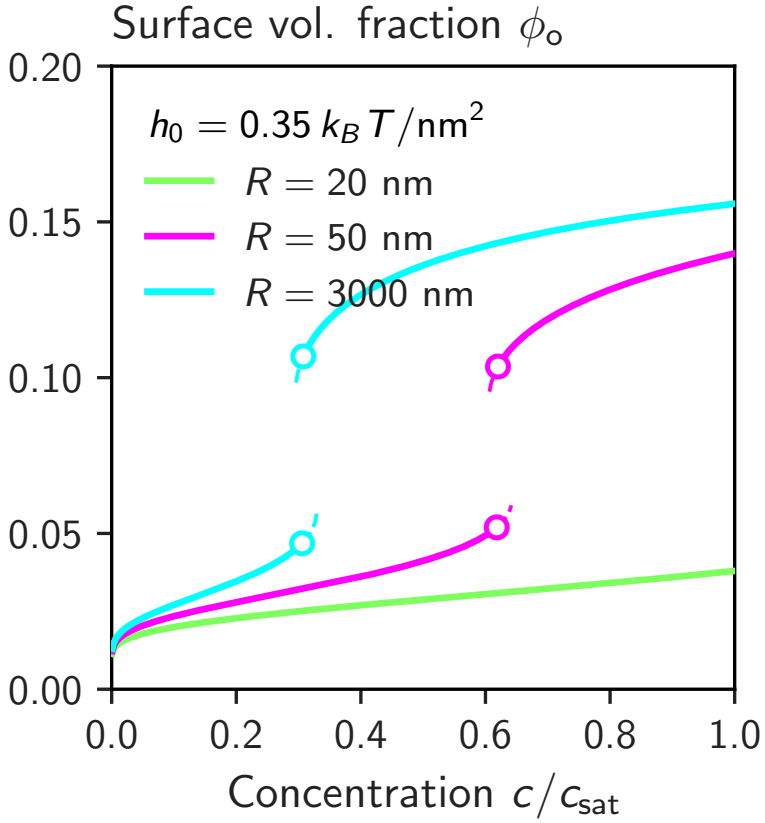
varying h_0



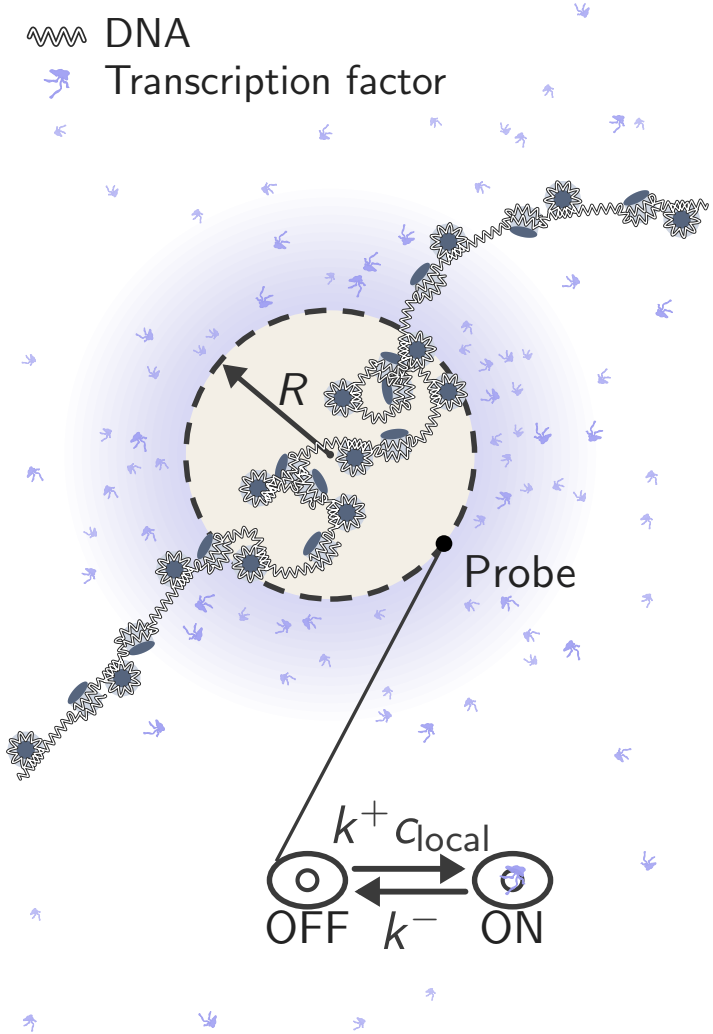
scene25_2



varying R



scene25$_3$



Free probe

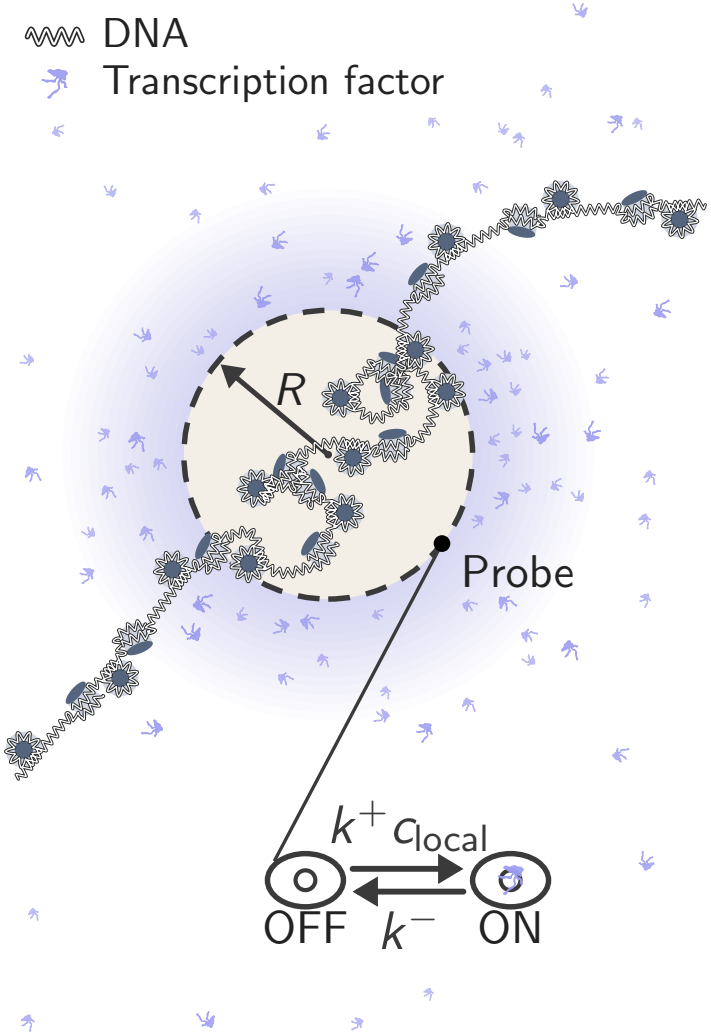
$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_r(c)}$$

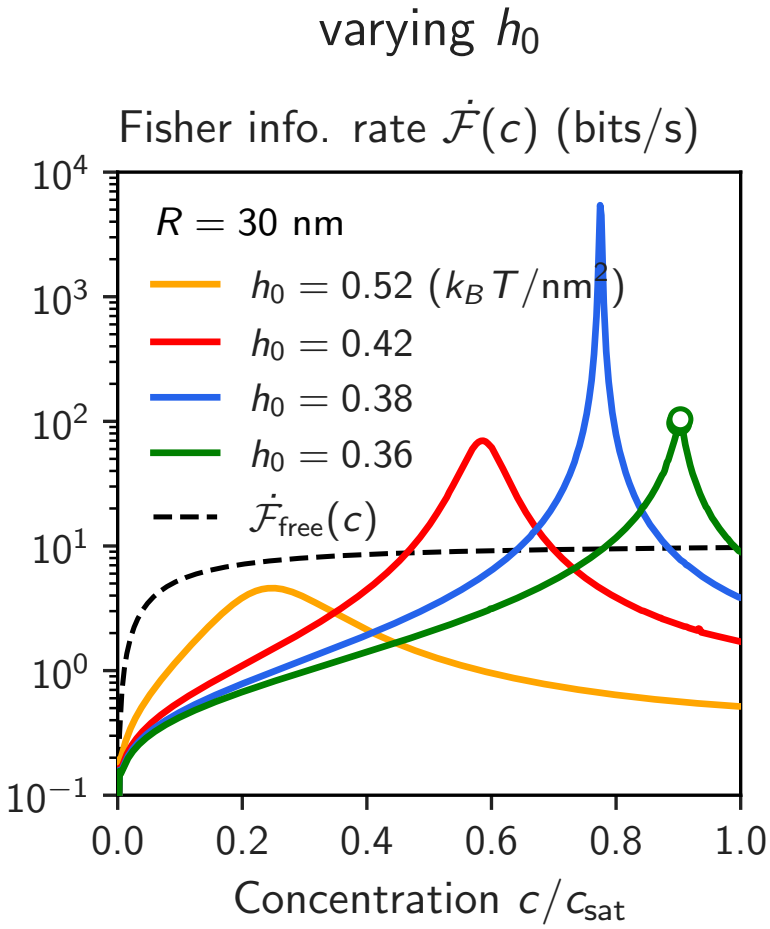
Probe on the anchor

$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_r(\phi)}}_{\text{probe}} \underbrace{\left(\frac{d \log \phi}{d \log c}\right)^2}_{\text{surface}}$$

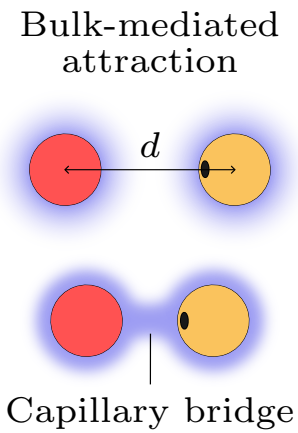
τ_r mean cycle time of the probe
ϕ TF volume fraction at the probe

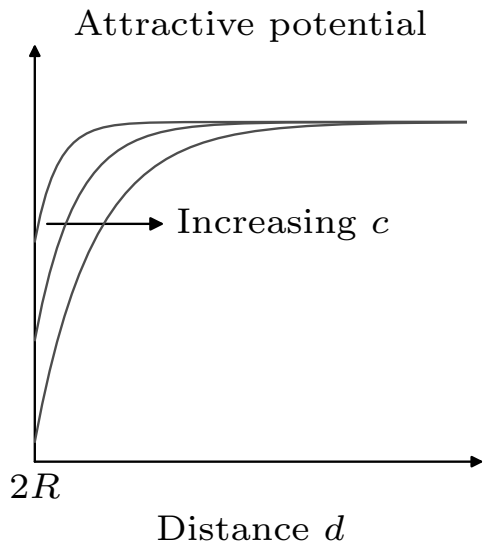
scene26$_1$

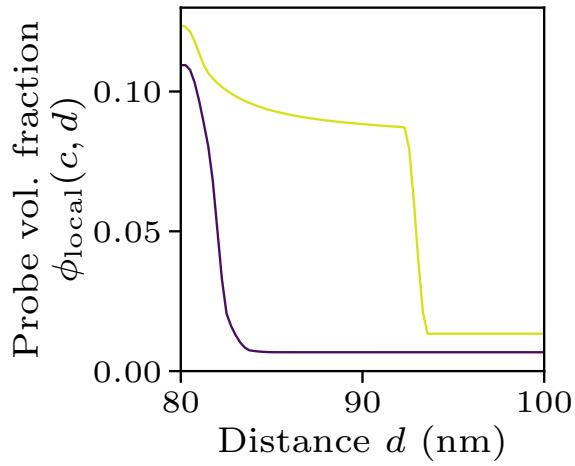

DNA
Transcription factor
R
Probe
$k^+ c_{\text{local}}$
OFF
k^-
ON

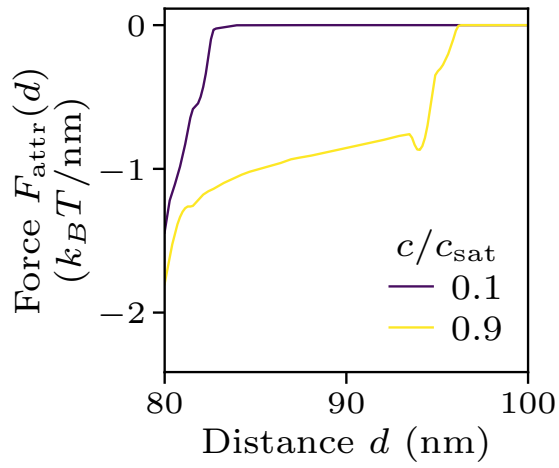

varying h_0
Fisher info. rate $\dot{\mathcal{F}}(c)$ (bits/s)
10^4
10^3
10^2
10^1
10^0
10^{-1}
$R = 30$ nm
$h_0 = 0.52$ ($k_B T$/nm^2)
$h_0 = 0.42$
$h_0 = 0.38$
$h_0 = 0.36$
$\dot{\mathcal{F}}_{\text{free}}(c)$
0.0
0.2
0.4
0.6
0.8
1.0
Concentration c/c_{sat}

scene26$_2$

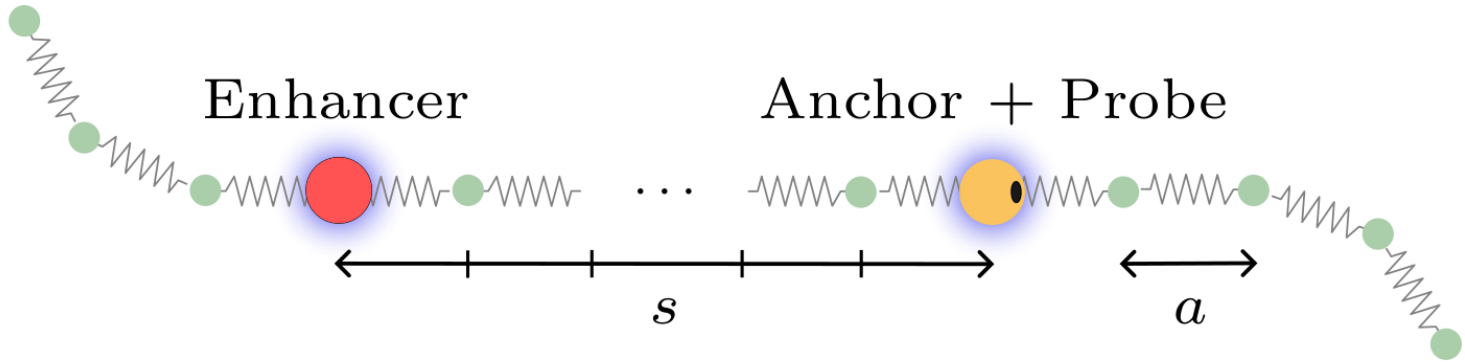

Bulk-mediated
attraction
d
Capillary bridge


Attractive potential
Increasing c
$2R$
Distance d

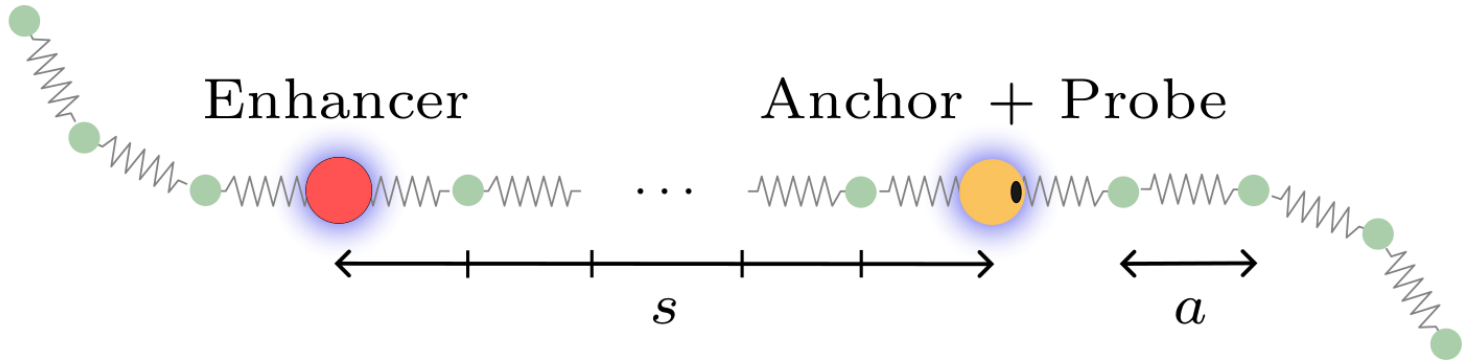

Probe vol. fraction
$\phi_{\text{local}}(c, d)$
0.10
0.05
0.00
80
90
100
Distance d (nm)


Force $F_{\text{attr}}(d)$
$(k_B T/\text{nm})$
0
−1
−2
c/c_{sat}
0.1
0.9
80
90
100
Distance d (nm)

scene27

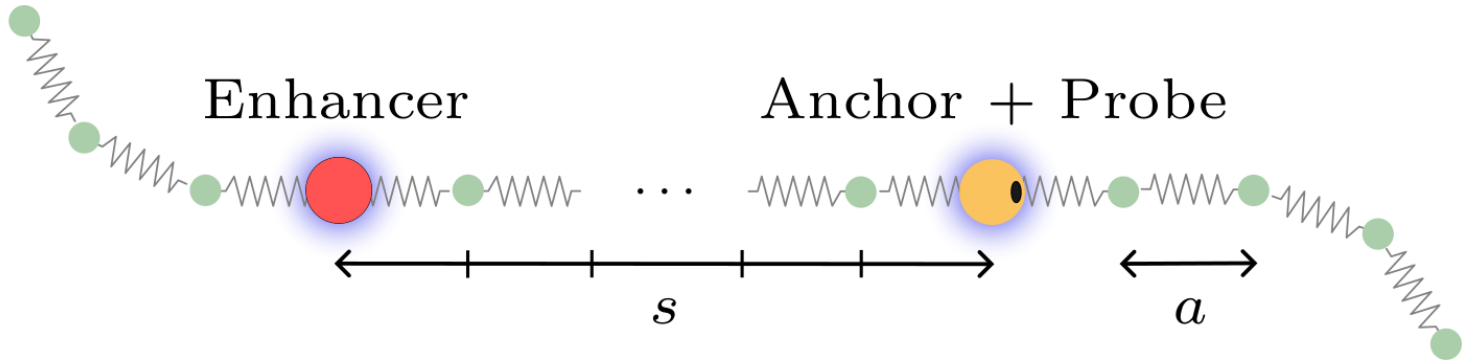


$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$



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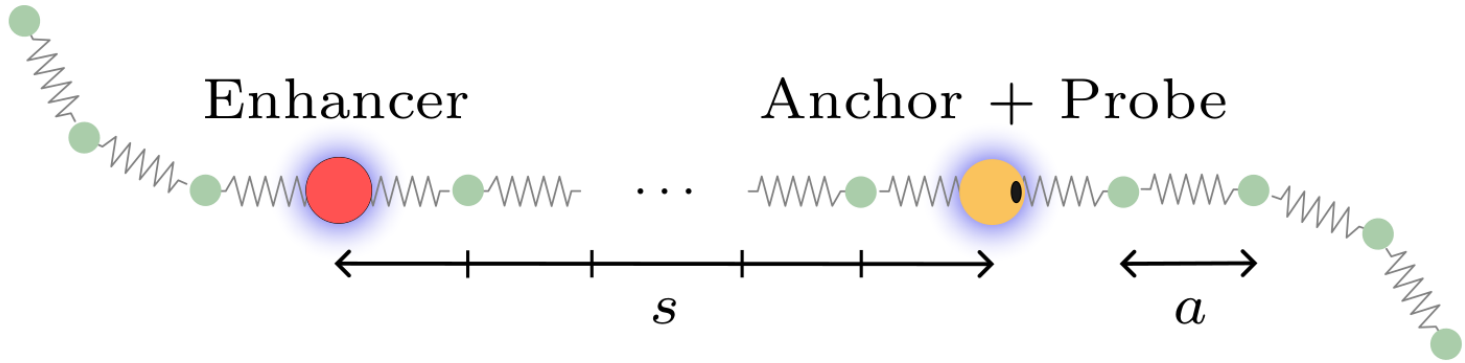
$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

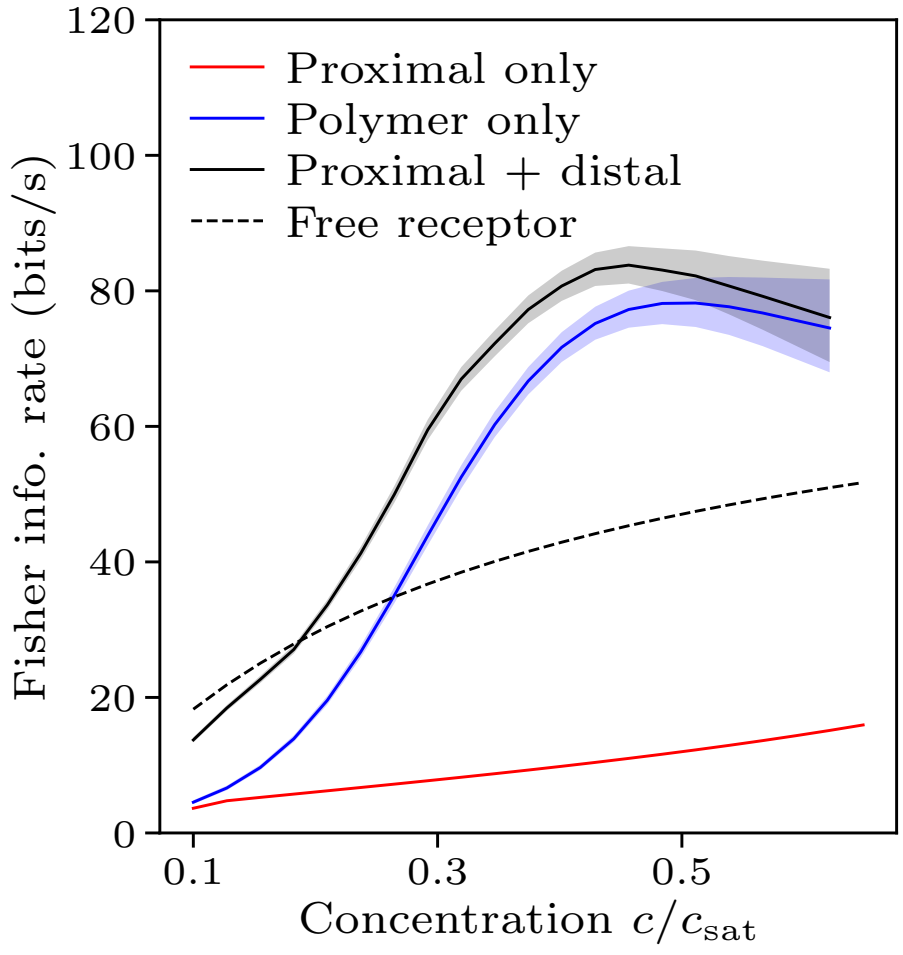


$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

a: Kuhn bond length
s: genomic separation
d: euclidean distance


Proximal only
Polymer only
Proximal + distal
Free receptor
Fisher info. rate (bits/s)
120
100
80
60
40
20
0
0.1
0.3
0.5
Concentration c/c_{sat}

scene30

Conclusions

- Optimization forces us to think about constraints

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- Optimization forces us to think about constraints
- Different constraints can give rise to qualitatively different behavior
- Evolution need not optimize individual components: sensing is a distributed process
- Averages don't tell the whole story: single microscopic events can be extremely informative